

Building Holographic Entanglement by Measurement

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Midjourney-S



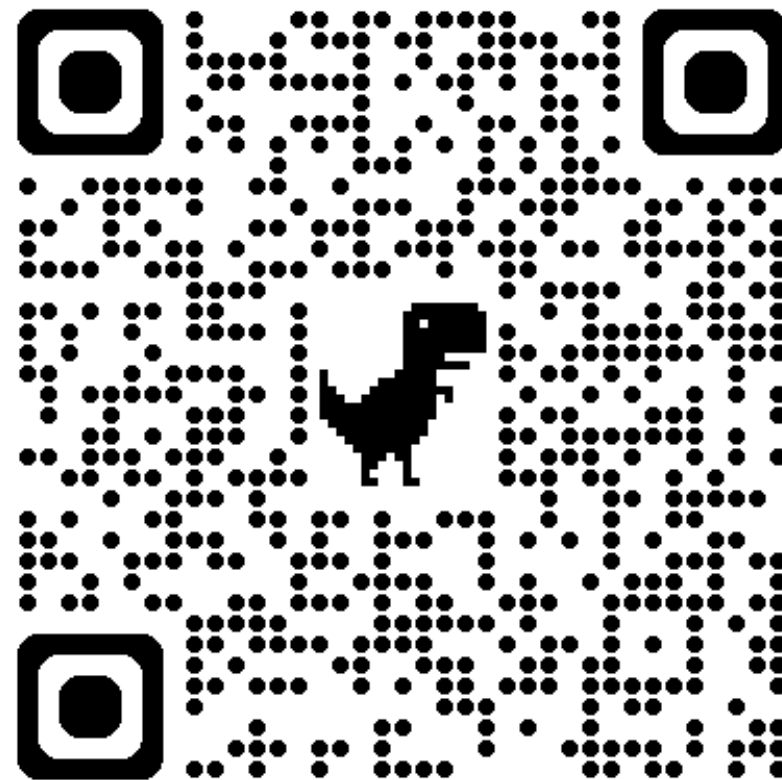
Jonathan Jeffrey



Monika Schleier-Smith

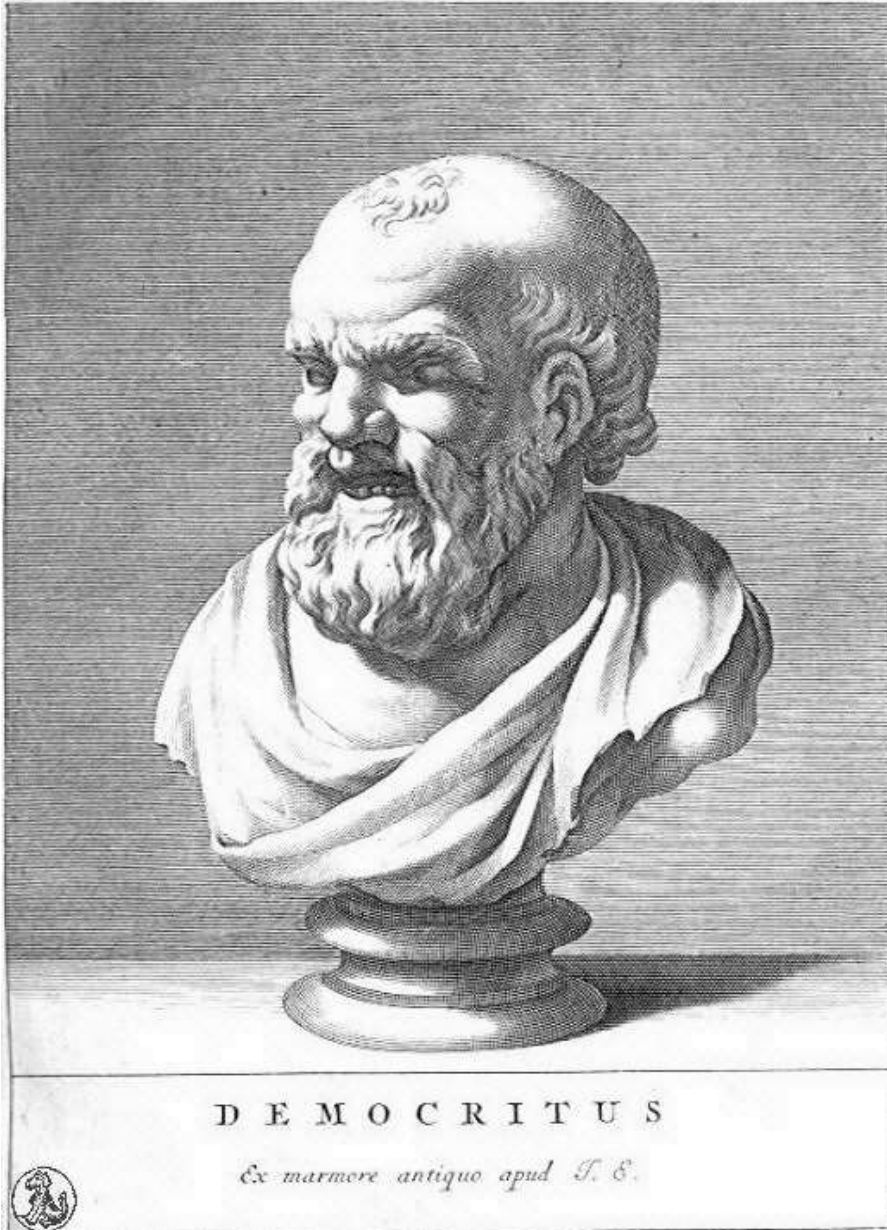


Lucien Gandarias



physicsmonkey.github.io

The problem of quantum gravity



“By convention sweet and by convention bitter, by convention hot, by convention cold, by convention color; but in reality atoms and void.”

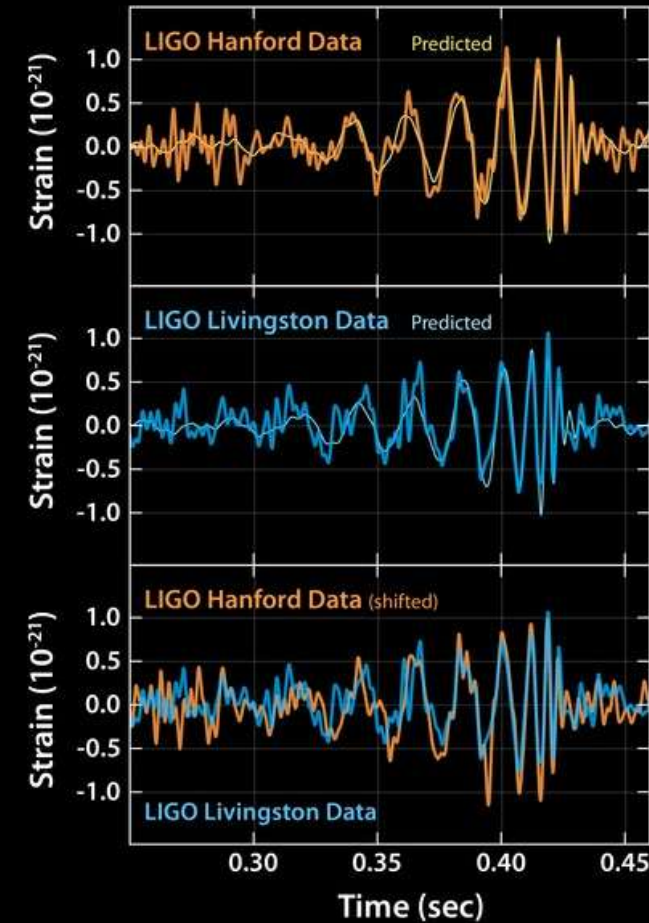
Why is the void interesting?

Quantum gravity is mysterious!

$$S = \frac{\text{Area}}{4G_N\hbar} \quad [\text{Bekenstein, Hawking}]$$



[Supermassive black hole in M87 – EHT, 2019]



$$\tau \sim \frac{\hbar}{T}$$

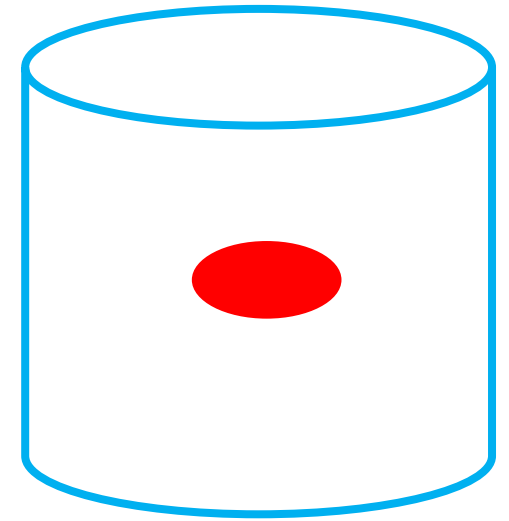
What are the “atoms” of spacetime and how do they yield the area/entropy relationship?

A toy model

AdS/CFT and “it from qubit”

For a “universe in a bottle” (Anti de Sitter space), the AdS/CFT correspondence gives a framework for quantum gravity: [Maldacena, ...]

- Wonderful theoretical lab to study black holes, strong coupling dynamics, hydrodynamics, ...
- Gravitational physics is intimately linked with quantum information: entanglement and geometry, complexity and black holes, ...
- By simulating the boundary system in a lab, we can hope to experimentally study holography!



Stuff in AdS $\leftrightarrow$
States in dual CFT

Big picture goal

Study models of holographic quantum gravity using table-top atomic, molecular, and optical (AMO) style experiments

- Asking how to do this, even just in principle, is illuminating
- Working within a specific experimentally motivated set of constraints is stimulating
- Ultimately, we want to carry out the experiments and learn!

Examples: lattice QFT, classical chaos, scrambling, ...

Clues: holographic nature of gravity

1. The Hamiltonian is measurable at infinity

$$F = \frac{G m_{\text{test}}}{r^2} \left(\frac{E}{c^2} \right) \quad [\text{Newton}]$$

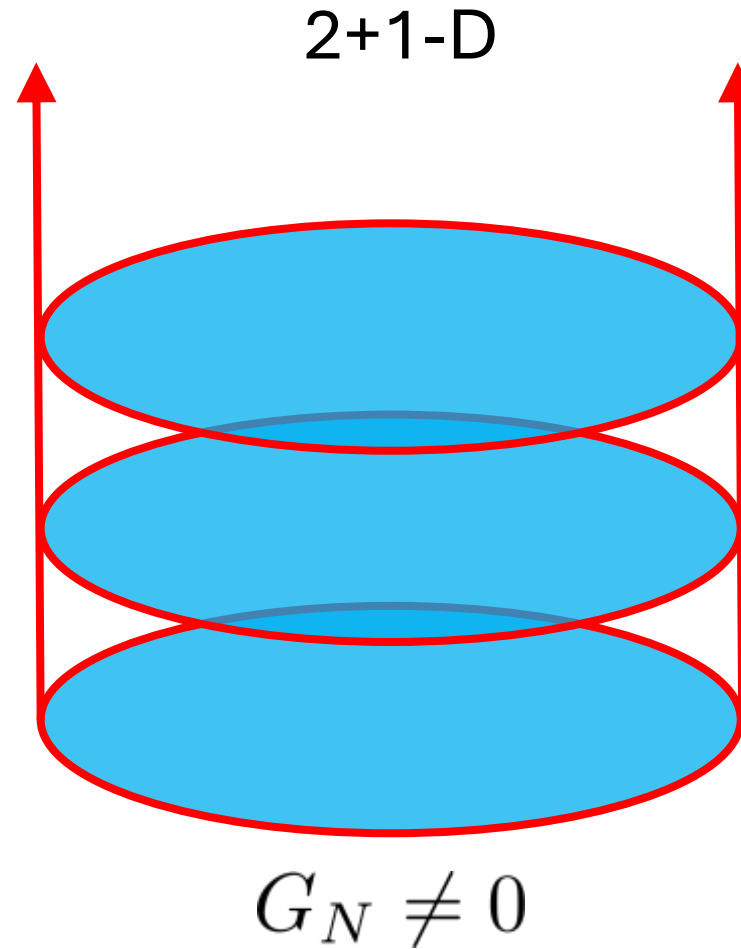
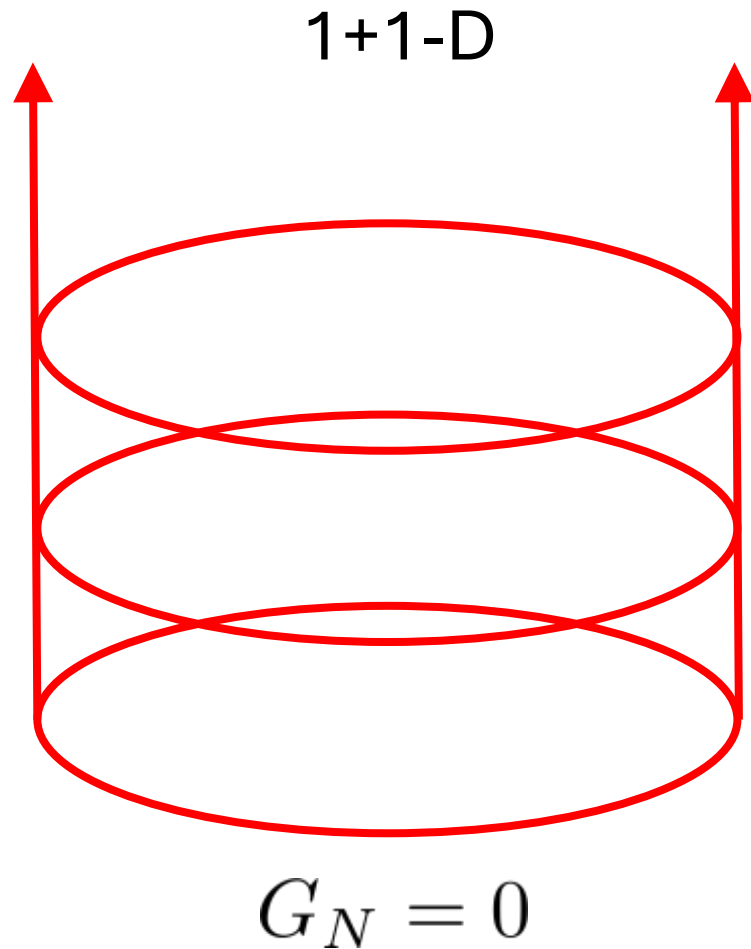
2. Black holes have entropy proportional to their area

$$S_{BH} = \frac{c^3 A_{BH}}{4\hbar G_N} \quad [\text{Bekenstein-Hawking}]$$

3. Gravitational redshift

$$E_{\infty} = E_0 \sqrt{1 - \frac{2GM}{rc^2}} \quad [\text{Einstein}]$$

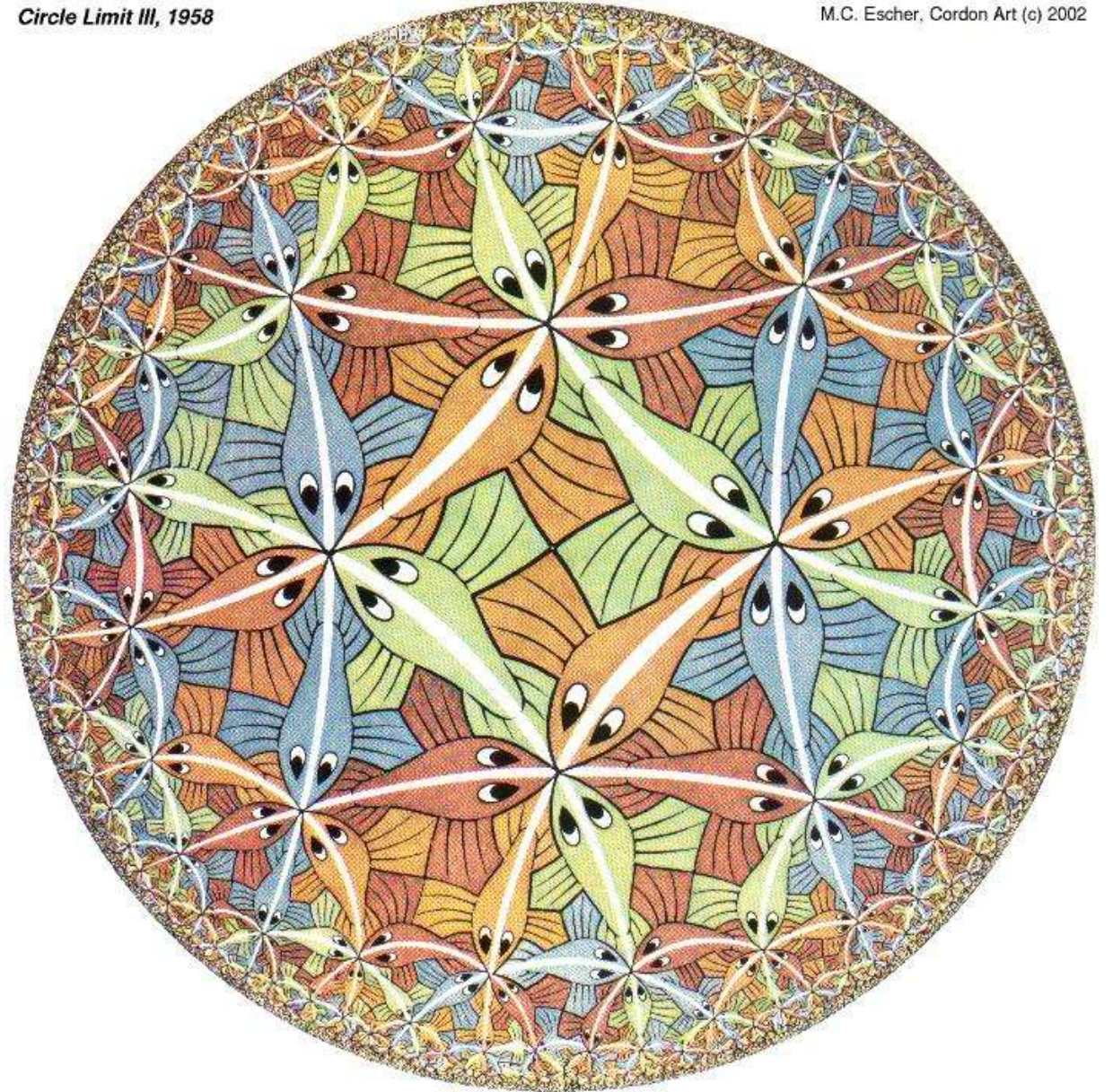
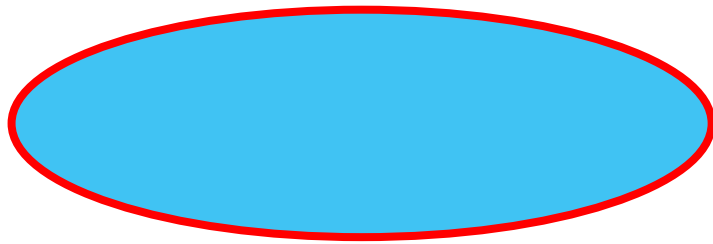
Anti de Sitter space / conformal field theory (AdS/CFT or holographic duality)



[Maldacena]

Circle Limit III, 1958

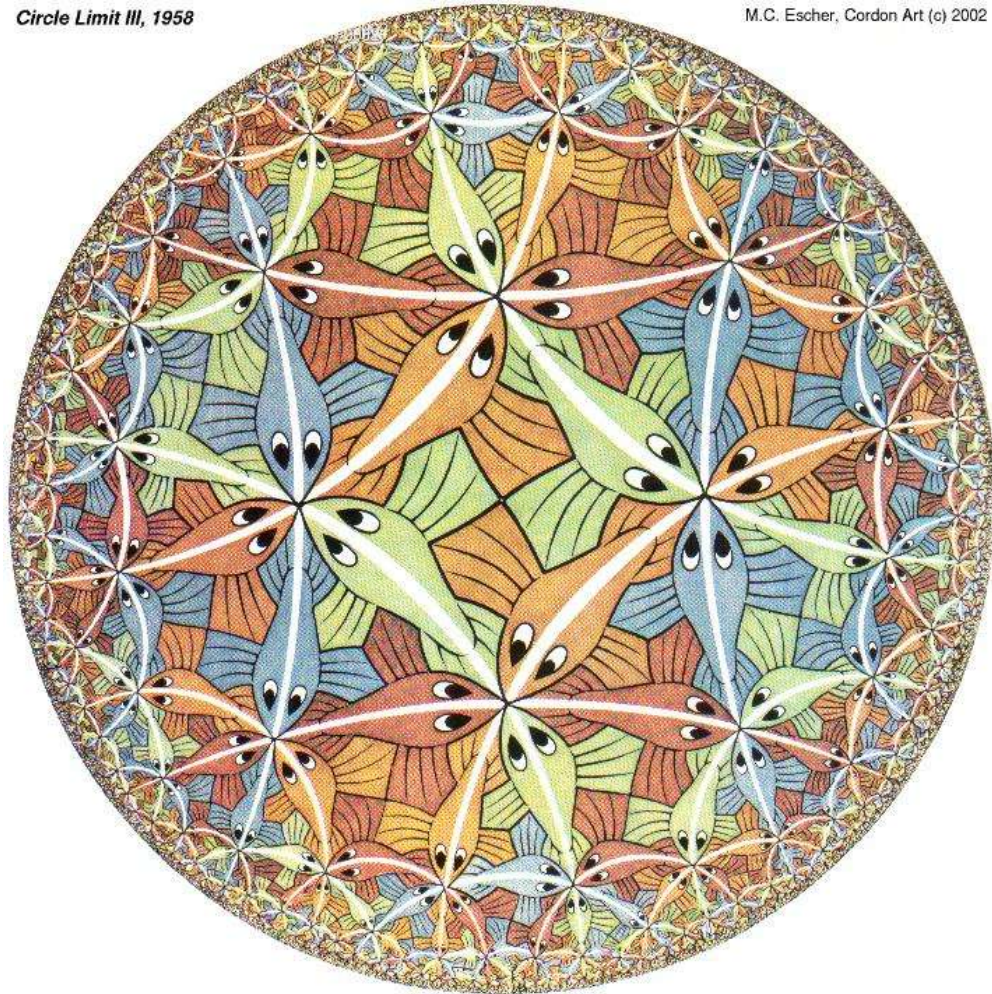
M.C. Escher, Cordon Art (c) 2002



AdS and an AdS black hole

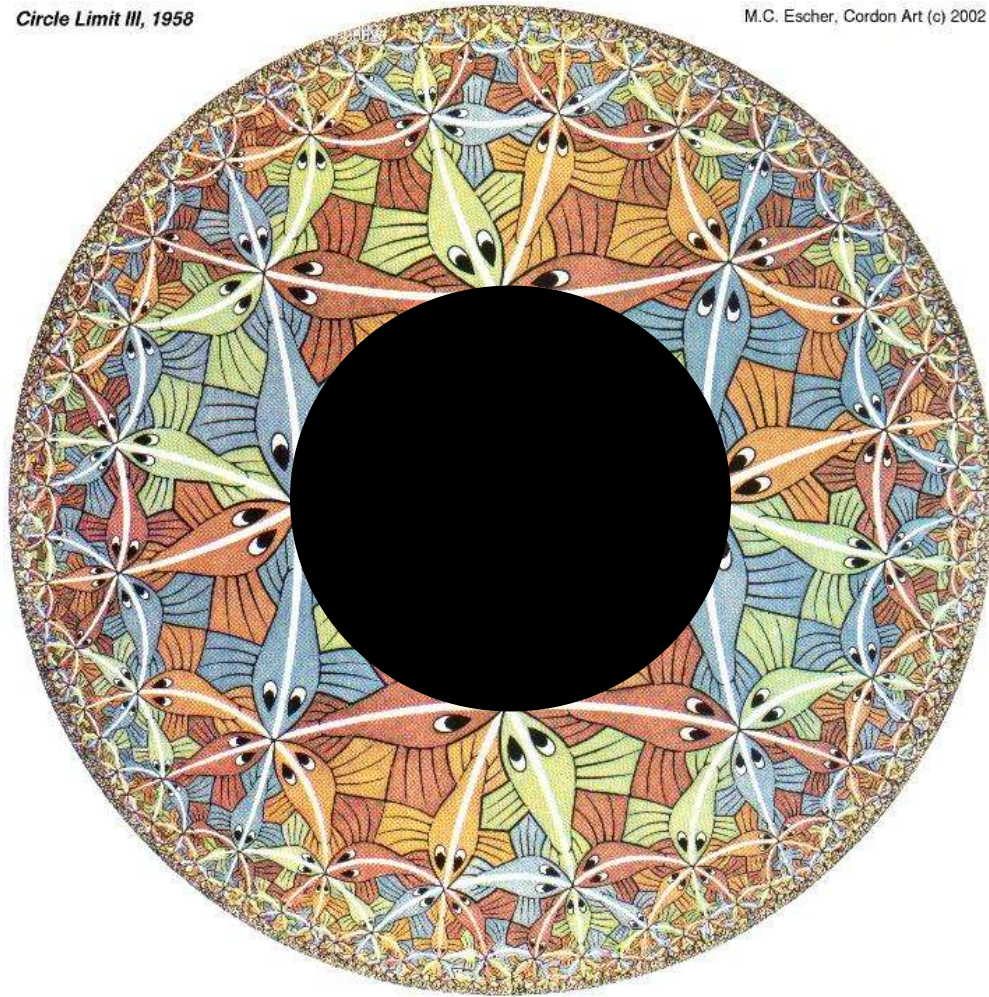
Circle Limit III, 1958

M.C. Escher, Cordon Art (c) 2002



Circle Limit III, 1958

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Entanglement

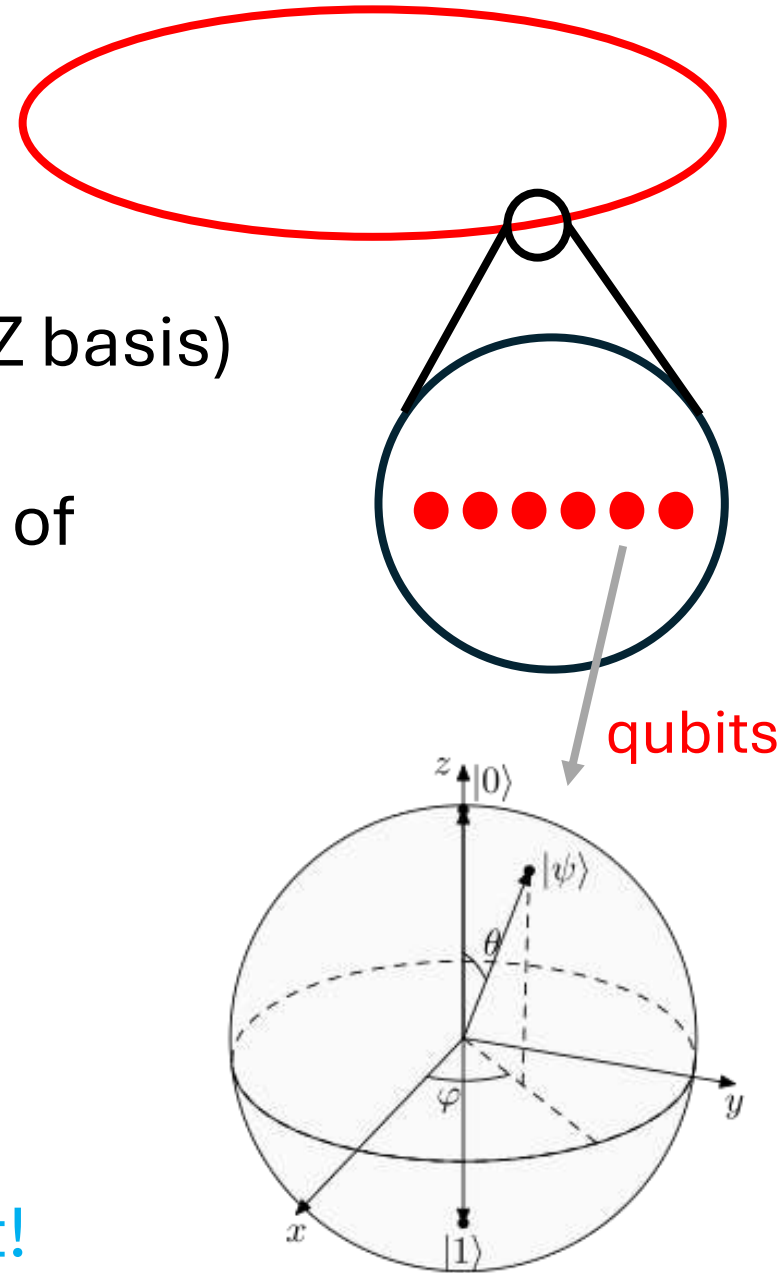
Four states: 00 and 01 and 10 and 11 (Z basis)

Superposition: any linear combination of states is also allowed

$$|\psi\rangle = \frac{|00\rangle + |11\rangle}{\sqrt{2}} \neq |\phi_1\rangle |\phi_2\rangle$$

EPR or Bell pair

Definition of entanglement!



Ryu-Takayanagi entanglement formula

S = Von Neumann entropy

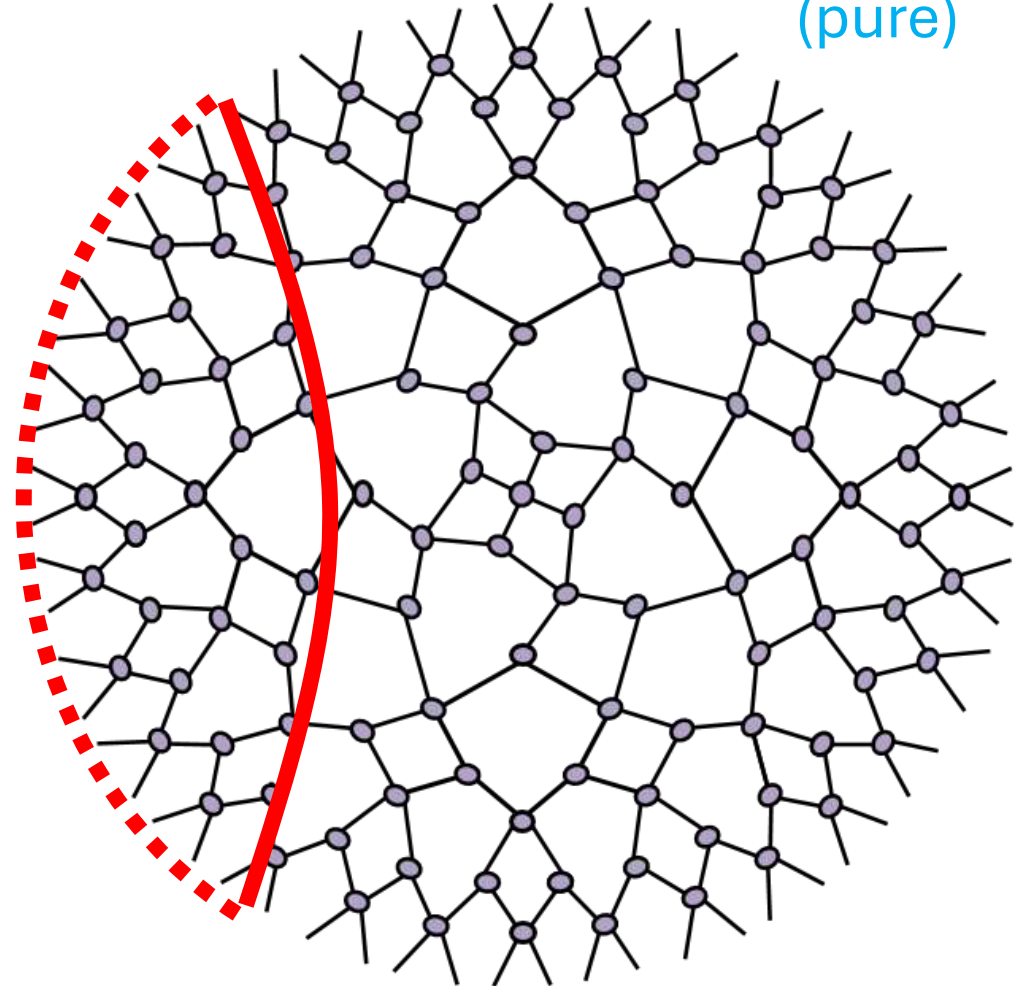
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A

$$S(A) = \frac{\text{Minimal Area}}{4\hbar G_N}$$

[Ryu-Takayanagi]

ground state
(pure)



[S, Pastawski et al., Hayden et al., ...]

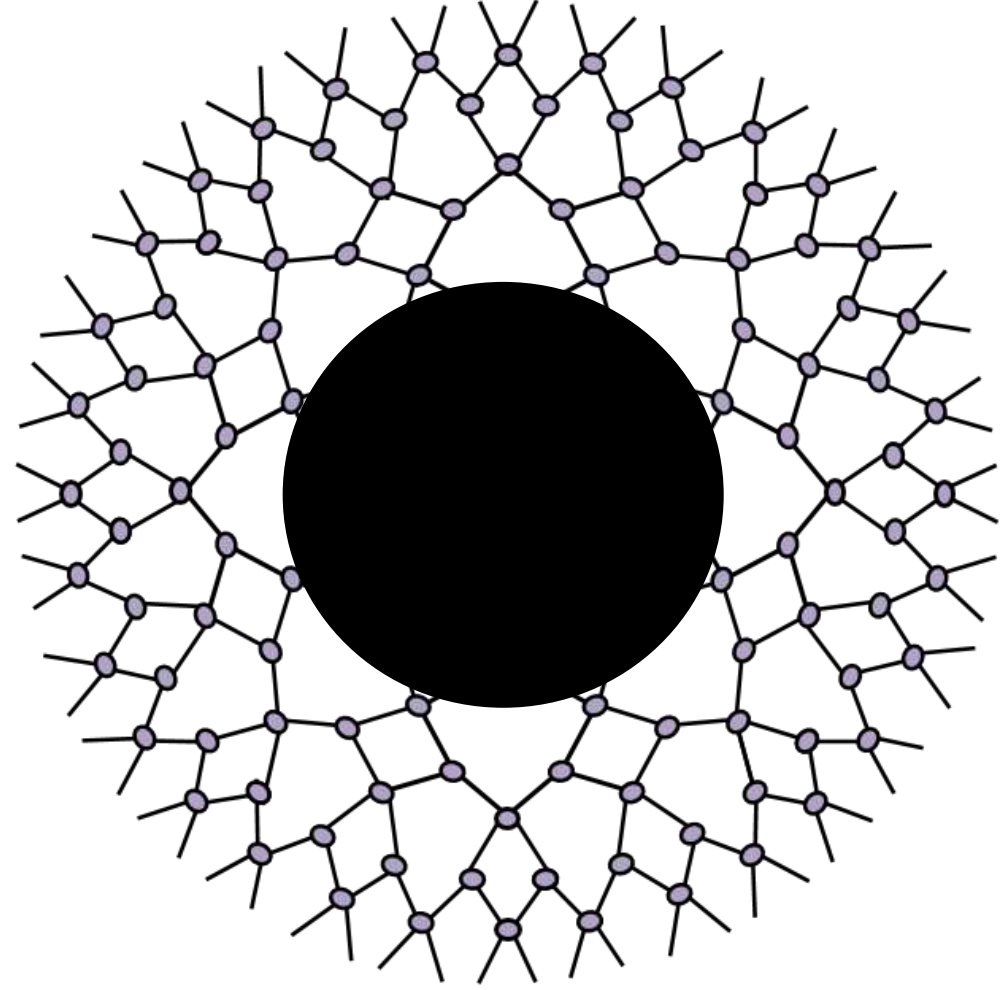
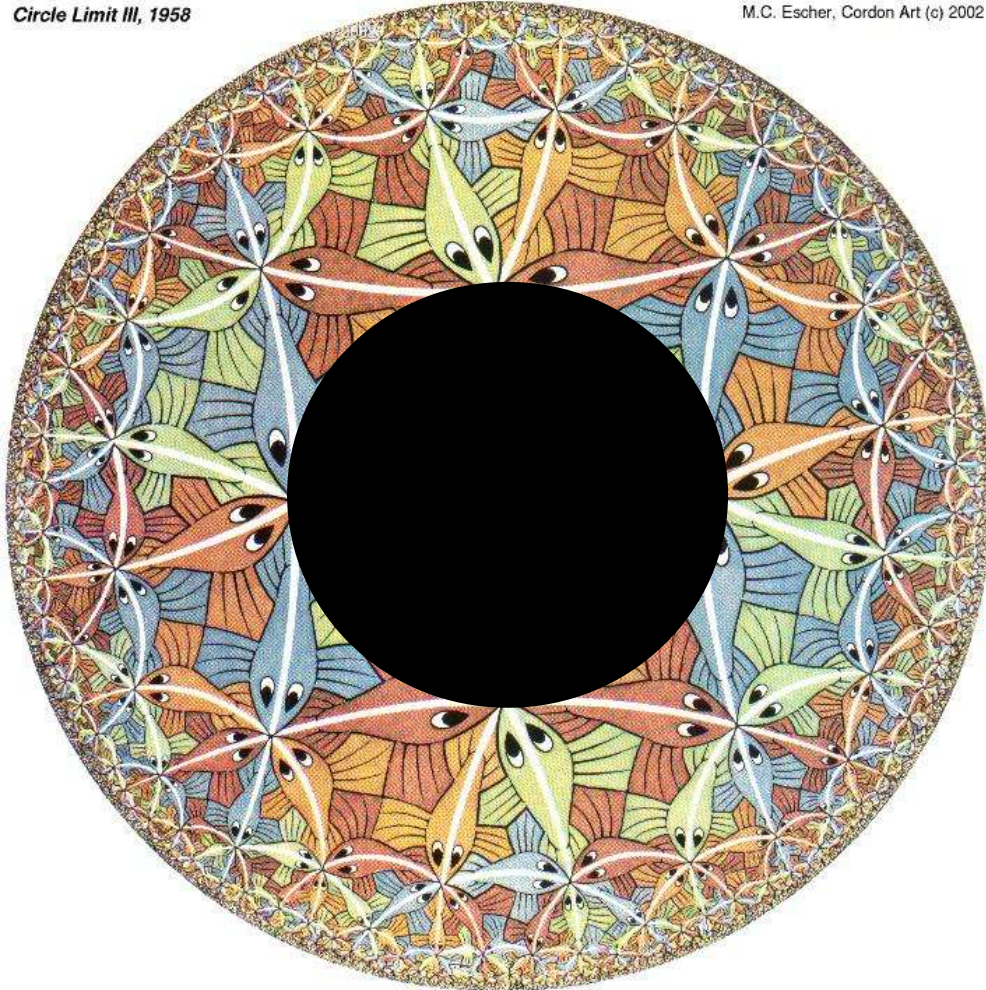
Building a black hole

$$\rho = \frac{e^{-\beta H}}{Z(\beta)}$$

thermal Gibbs
state (mixed)

Circle Limit III, 1958

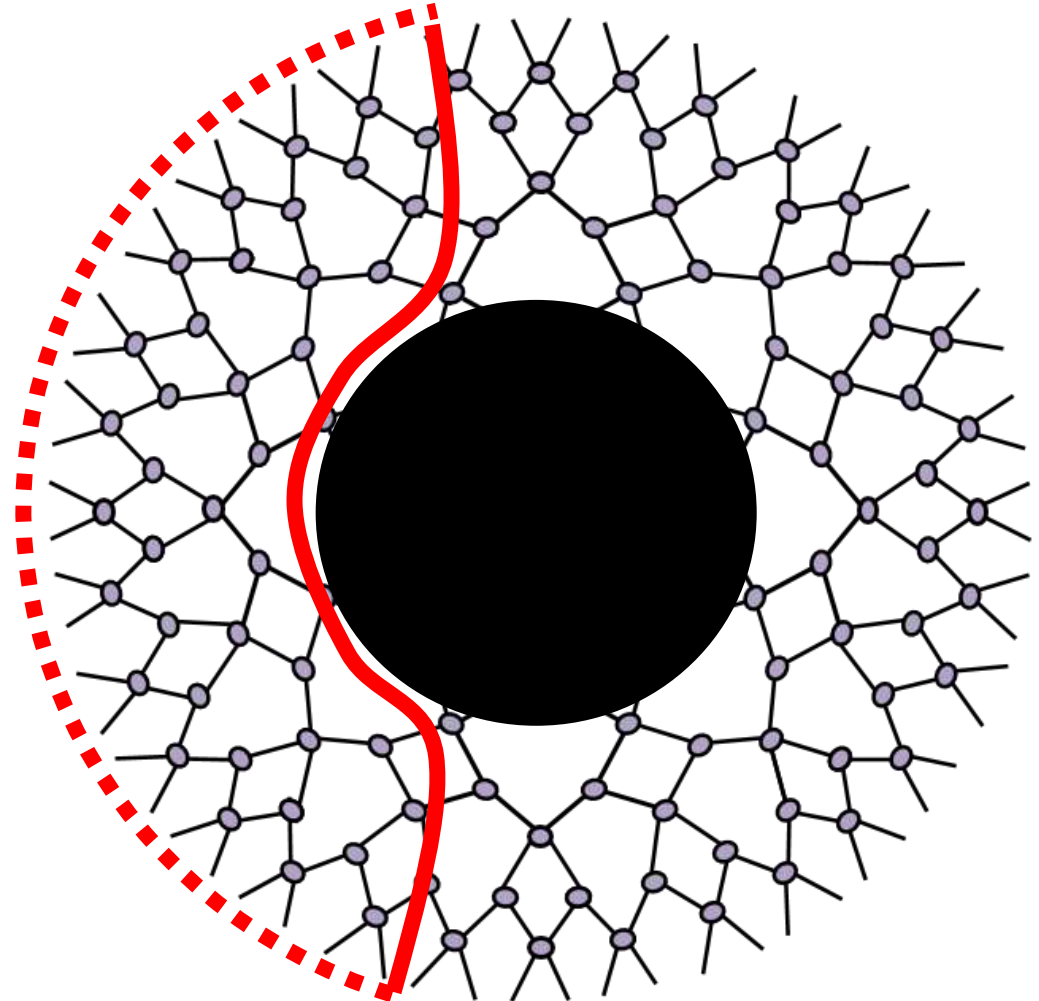
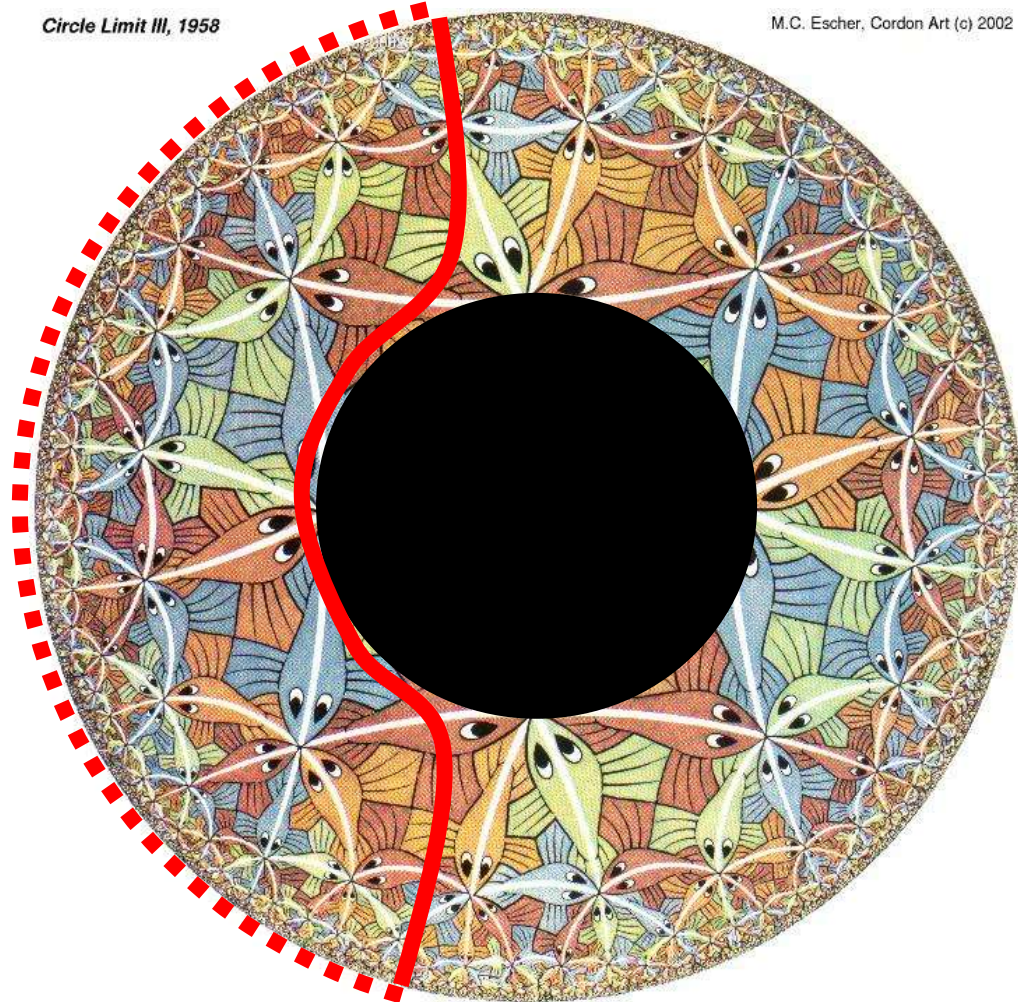
M.C. Escher, Cordon Art (c) 2002



Black hole entropy

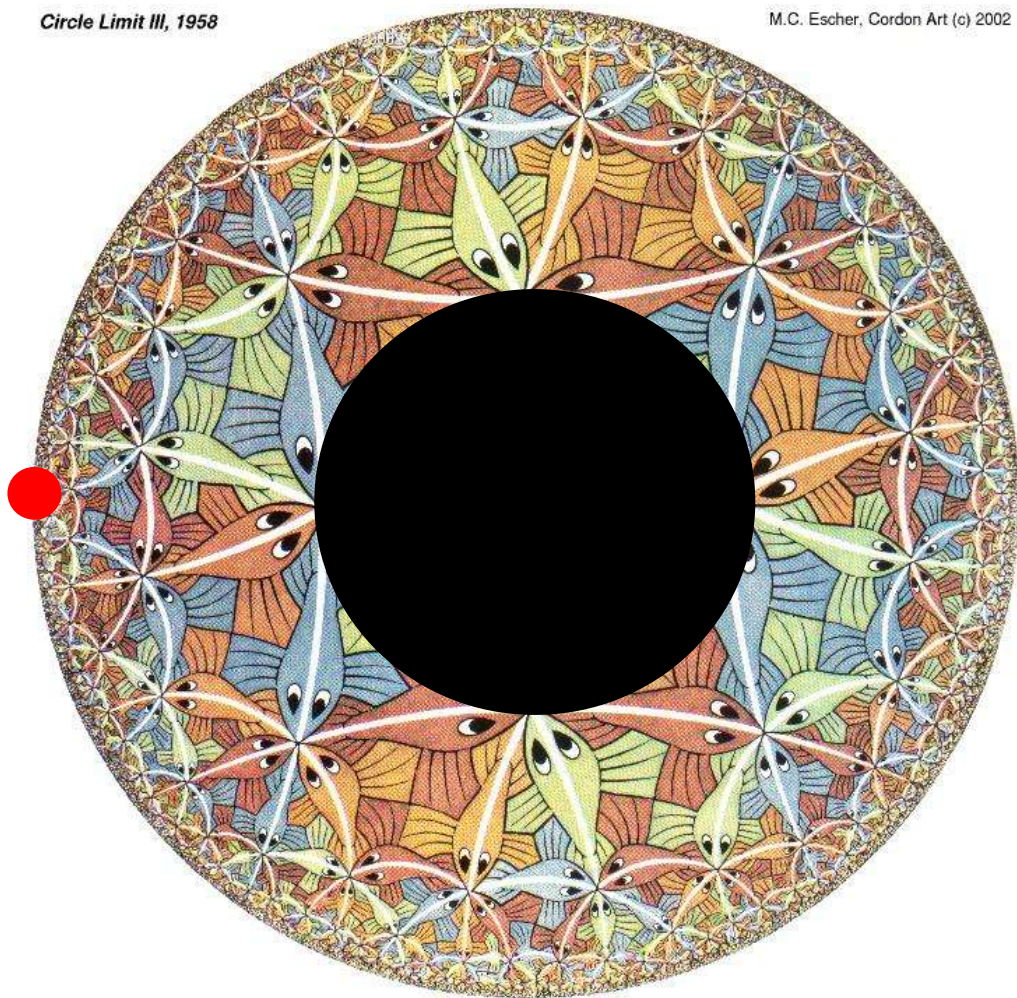
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M.C. Escher, Gorden Art (c) 2002



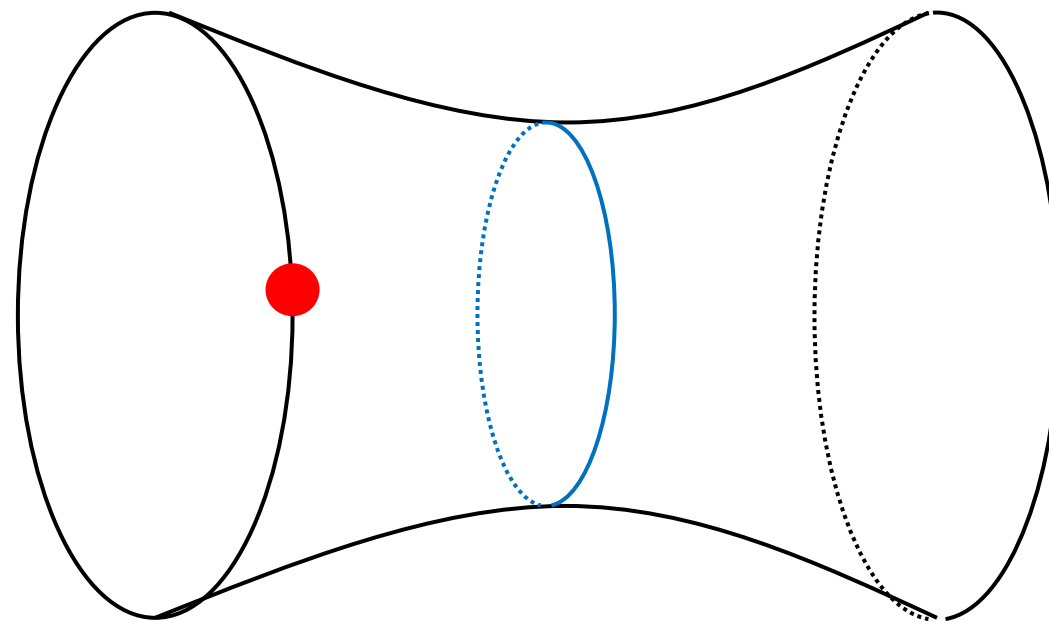
Circle Limit III, 1958

M.C. Escher, Cordon Art (c) 2002



~ mixed state

Two-sided wormhole

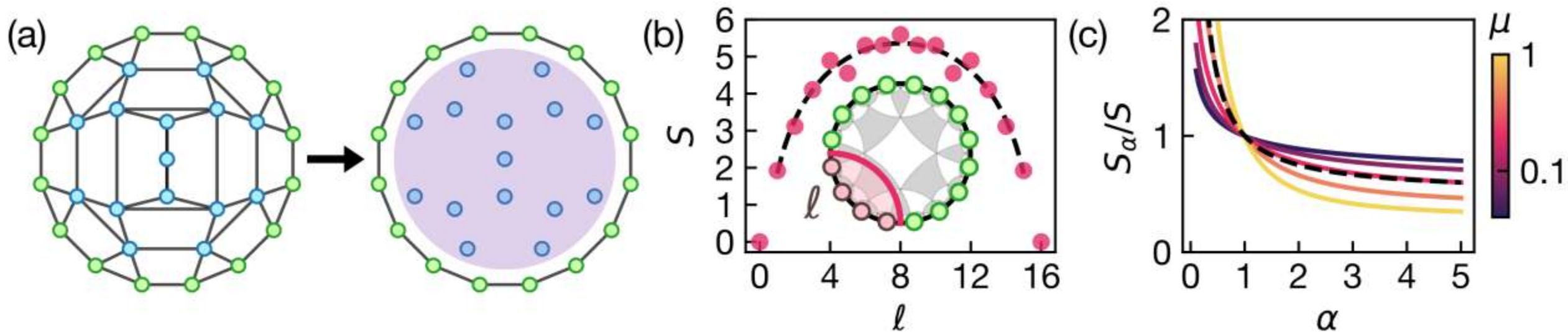


~ purification

Entanglement from geometry

Our proposal and key results

- Inspired by experimentally motivated constraints, we exhibit a simple setting with Ryu-Takayanagi-like (RT) entanglement
- Start with a graph which discretizes a chosen geometry, quench for $O(1)$ time, and measure the bulk $\rightarrow$ a pure boundary state



Key ingredients

- Degrees of freedom are sites that each contain a single canonical pair (x,p) {for example, each site may be a collection of atoms with a collective angular momentum, $[S_x, S_y] = i\hbar^2 s + i\hbar(S_z - s\hbar)$ }
- Sites have arbitrary interaction connectivity {this could be achieved by coupling atomic ensembles with light}
- Any Gaussian operations are allowed: phase space shifts, evolution with quadratic Hamiltonians, and measurements of x,p
- These ingredients are motivated with atomic ensembles in a cavity in mind, but there are other experimental platforms, e.g. photons

Setup

- The talk assumes the following protocol:
 1. Choose a graph $G=(V,E)$ with bulk and boundary sites comprising V
 2. The initial wavefunction is a product of Gaussians for each site in V
 3. The system is quenched: evolve with $H_q = \frac{1}{2} \sum_{ab} x_a J_{ab} x_b$ for unit time where J is set by the (weighted) adjacency matrix of G
 4. Measure the bulk sites in the momentum basis, and (optionally) apply local phase space displacements conditioned on the outcomes
- We analyze the entanglement entropy, Renyi entropy, and correlations of the resulting pure states

$$\xi = (x_1, \dots, x_N, p_1, \dots, p_N) \quad V_{ij} = \frac{1}{2} \langle \{ \xi_i - \langle \xi_i \rangle, \xi_j - \langle \xi_j \rangle \} \rangle$$

[literature: continuous variable graph states, methods to compute entropies]

$$e^{-\mu x_A^2/2 - \mu x_B^2/2 - iJ x_A x_B}$$



Highly squeezed limit

- Two modes (x_A, x_B) $\psi(x_A, x_B) \sim e^{iJ x_A x_B}$ highly squeezed, $\mu \ll 1$

$$\rho(x_A, x'_A) \sim \int dx_B e^{iJ x_A x_B - iJ x'_A x_B} \sim \delta(x_A - x'_A)$$

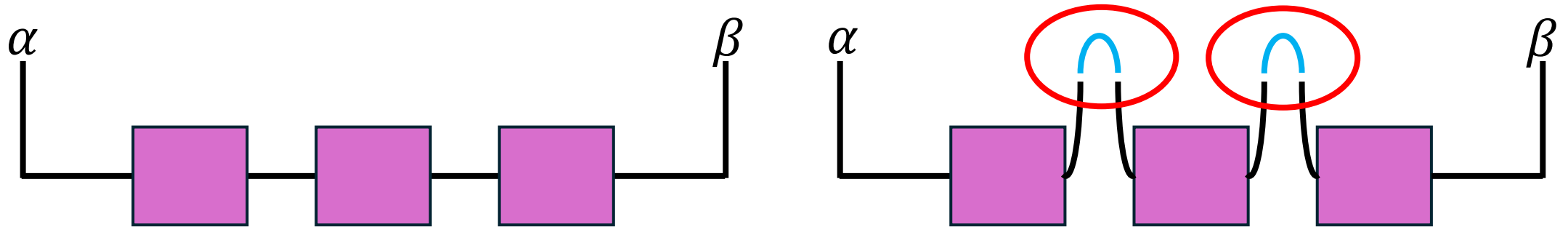
- k A-modes, k B-modes $\psi(x_A, x_B) \sim e^{\frac{i x_A J_{AA} x_A}{2} + \frac{i x_B J_{BB} x_B}{2} + i x_A J_{AB} x_B}$

$$\rho(x_A, x'_A) \sim \delta(x_A - x'_A) \text{ provided } J_{AB} \text{ is full rank}$$

[“perfect”-like tensors, see also [Kwon-Brady-Albert](#)]

Intuition

- Key inspirations: **tensor networks** and **measurements**
 - Tensor networks give a useful toy model for holography (e.g. recent MERA-based toy models of closed universe cosmology [[Antonini-Sasieta-S](#)])
 - Tensor networks also provide a powerful computational tool (e.g. TMERA [[Sewell-White-S](#)], forthcoming work on chiral clock model [[Guo-S](#)])
 - Measurements, when combined with feedforward, can yield efficient preparations of states (e.g. [literature on topological state prep](#))

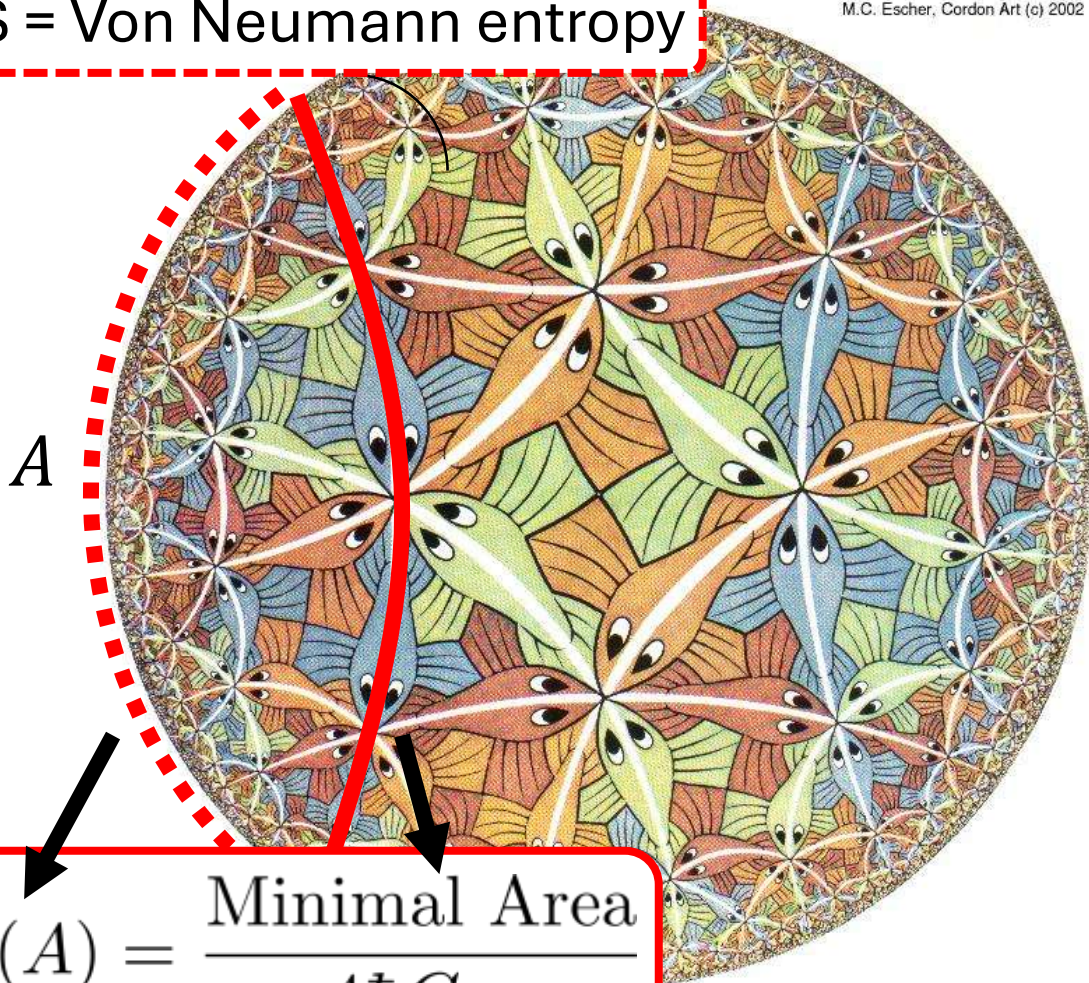


Implement internal contractions
by projecting onto Bell pairs

Example 1: hyperbolic disk

S = Von Neumann entropy

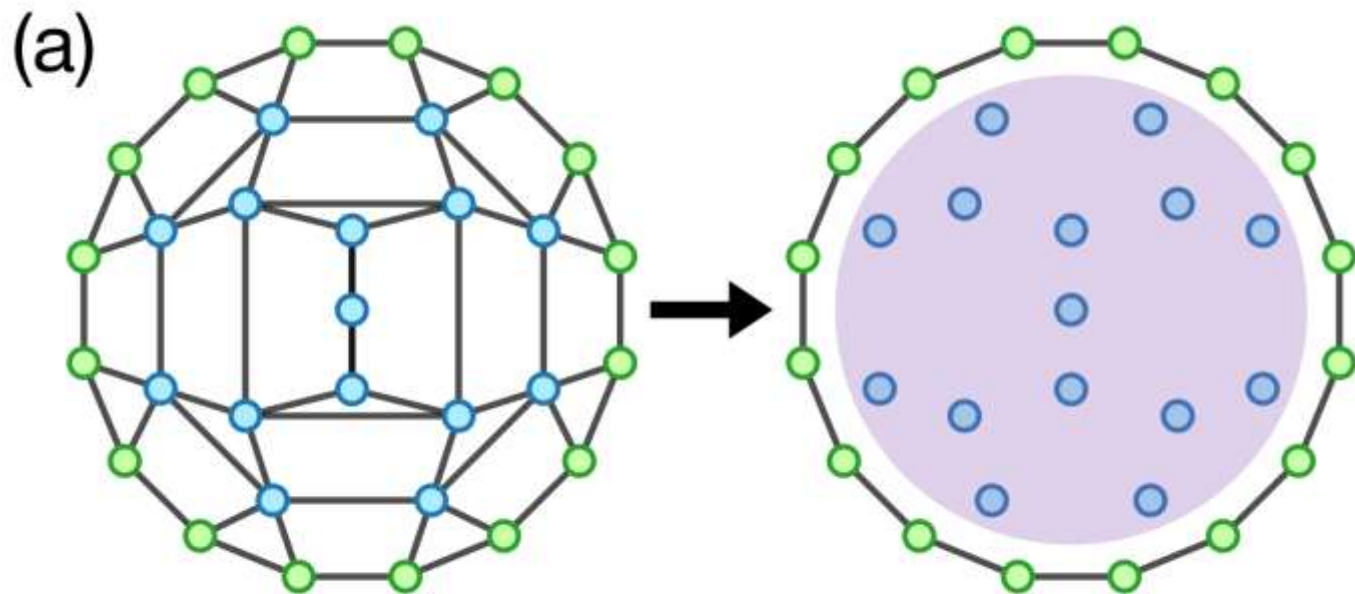
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- **AdS/CFT**: the ground state of the boundary (1+1)d CFT is dual to empty (2+1)d AdS space, spatial geometry is a hyperbolic disk
- The RT surface (curve in 2+1) is a portion of a circle (set $\hbar = 1$)

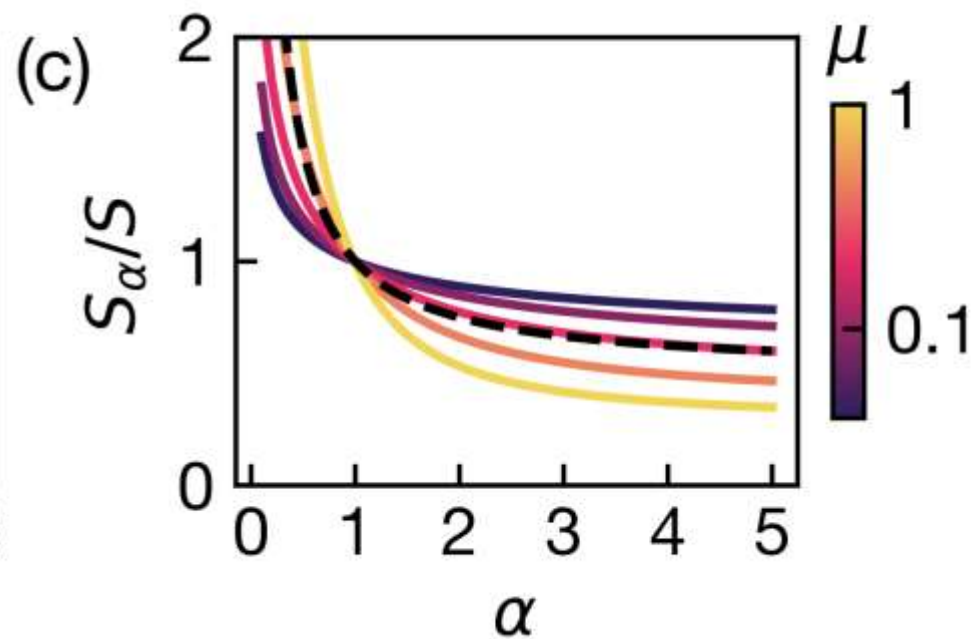
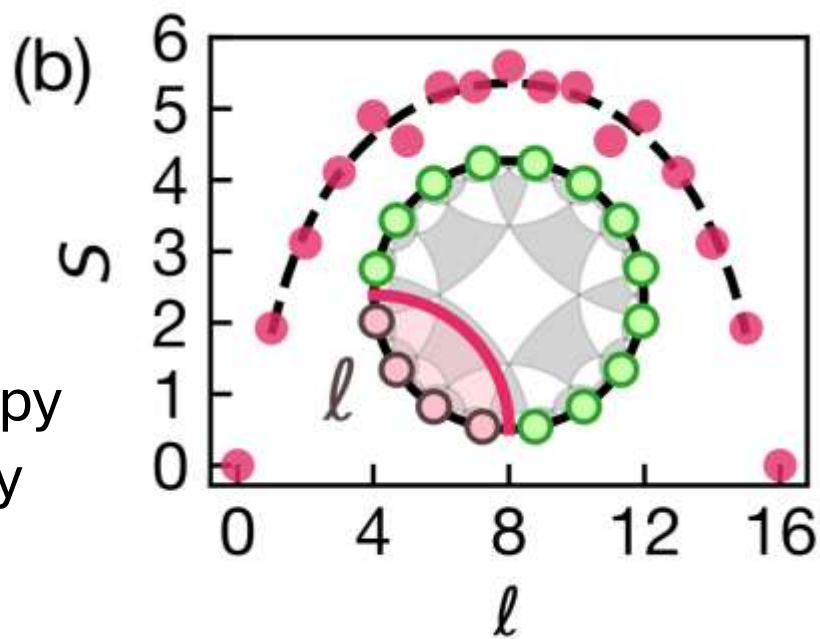
$$S = \frac{d_{\text{bulk}}}{4G} = \frac{\ell_{\text{AdS}}}{2G} \ln \left(\frac{2r_c}{\ell_{\text{AdS}}} \sin \frac{\Delta\theta}{2} \right)$$

$$S(A) = \frac{\text{Minimal Area}}{4\hbar G_N}$$



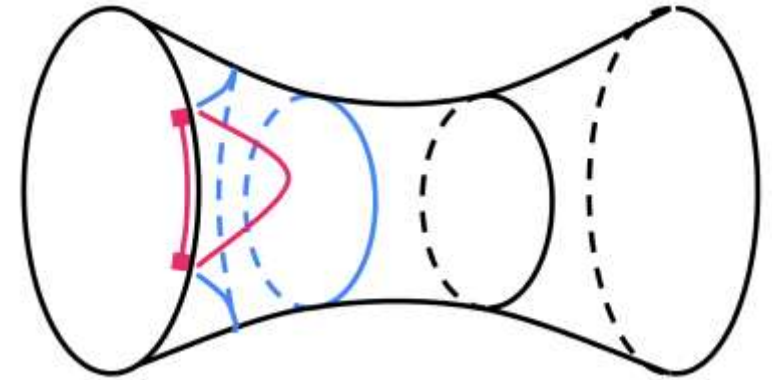
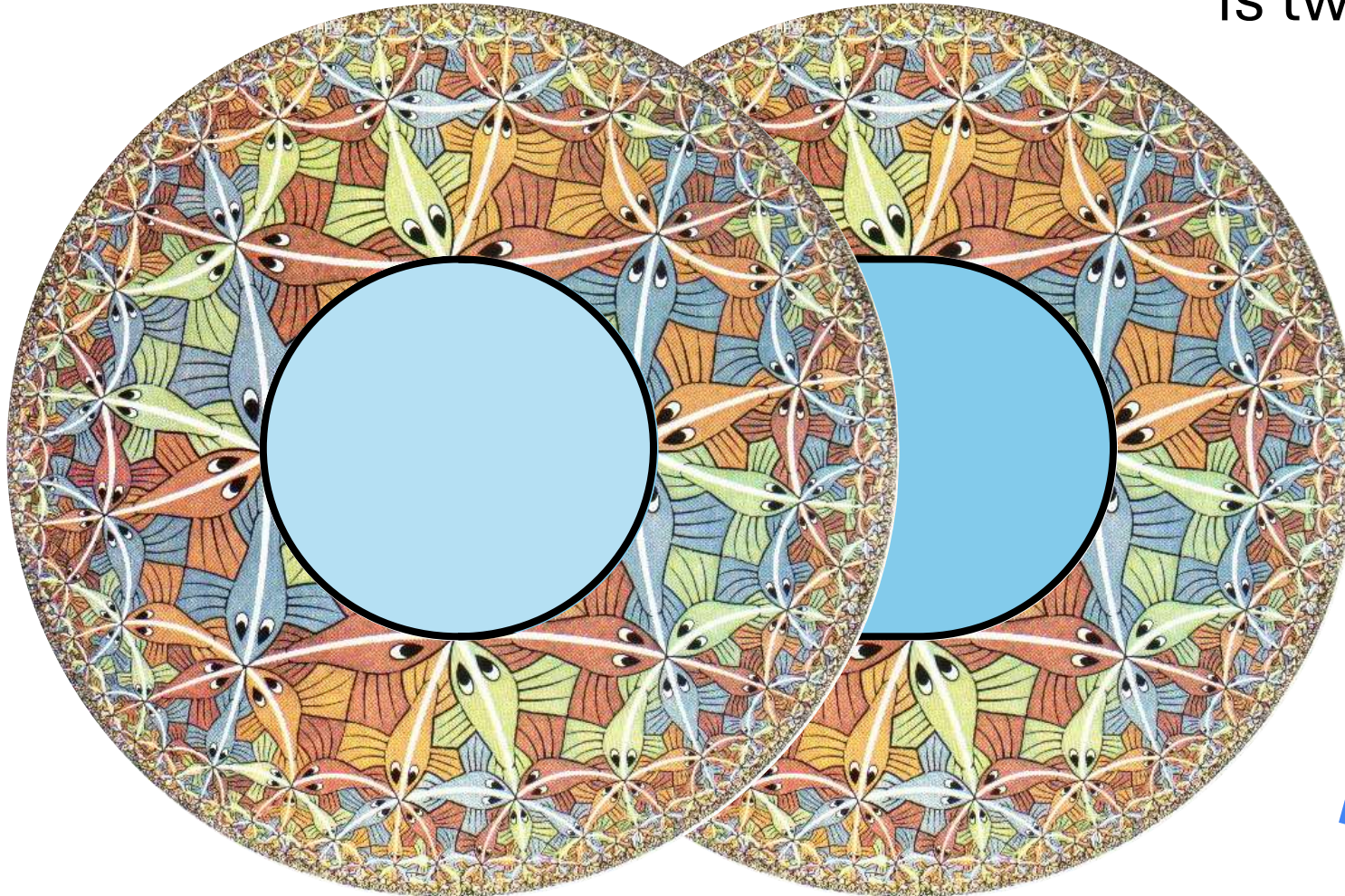
A simple discretization of the hyperbolic disk, followed by measurement of the bulk sites

(b) Entanglement entropy $\mu = .2$, (c) Renyi entropy as a function of μ



Example 2: wormhole

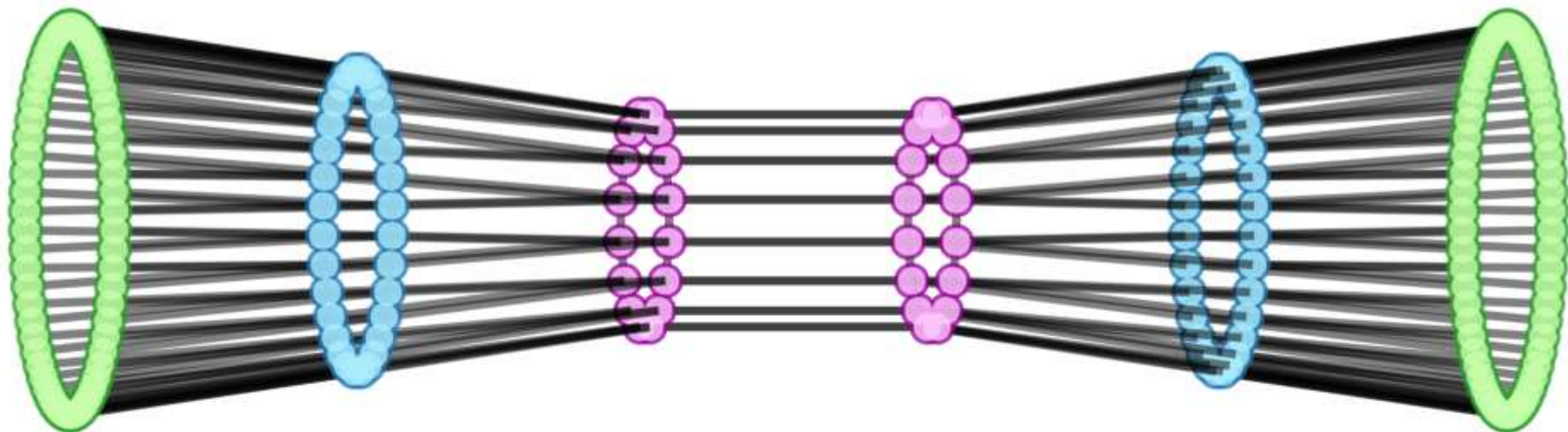
- **AdS/CFT**: a thermofield double state of the boundary (1+1)d CFT is dual to a wormhole connecting two black holes, spatial geometry is two annuli connected by a tube



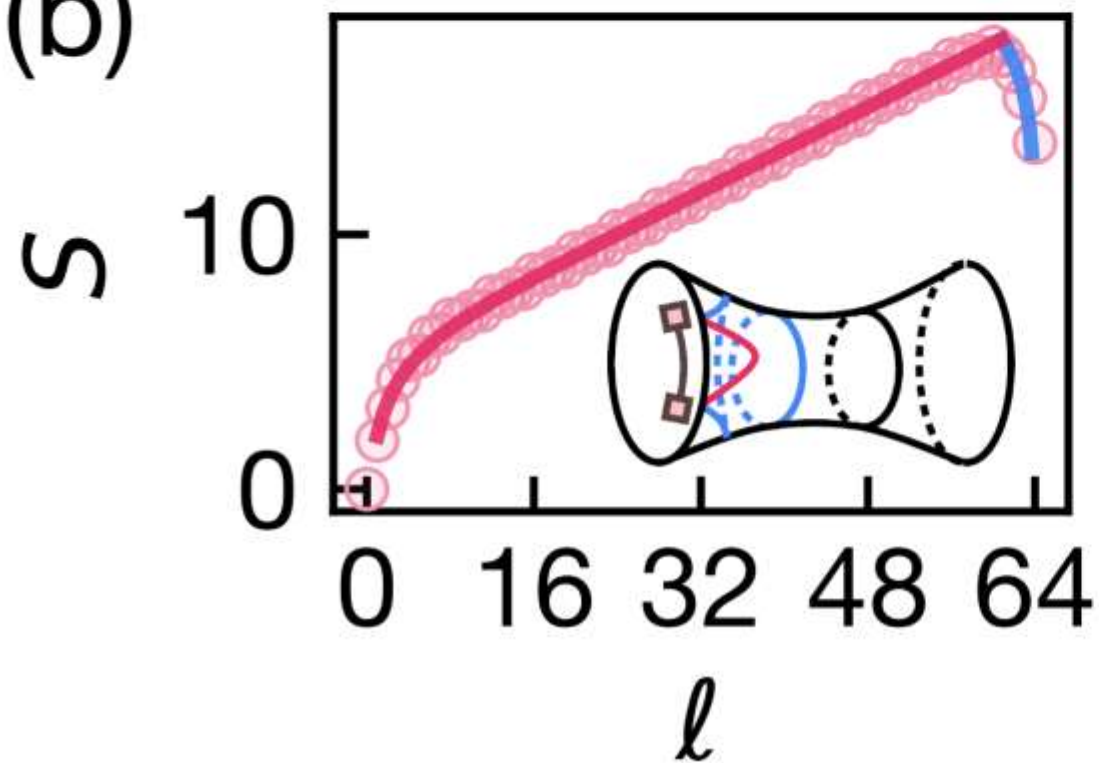
/
$$S^{(1)} = \frac{d_{\text{bulk}}}{4G} = \frac{c}{3} \ln \left(\frac{2r_c}{r_+} \sinh \frac{r_+ \Delta\theta}{2\ell_{\text{AdS}}} \right)$$

/
$$S^{(2)} = S_{th} + \frac{c}{3} \ln \left(\frac{2r_c}{r_+} \sinh \frac{r_+ (2\pi - \Delta\theta)}{2\ell_{\text{AdS}}} \right)$$

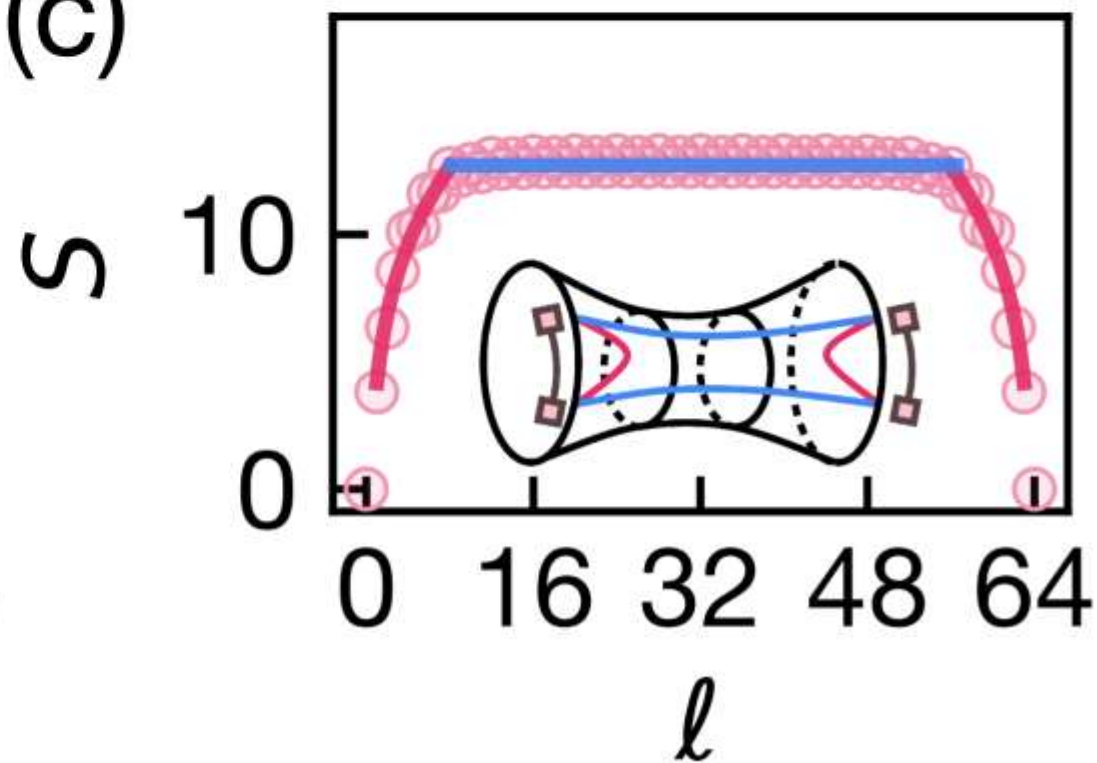
(a)



(b)

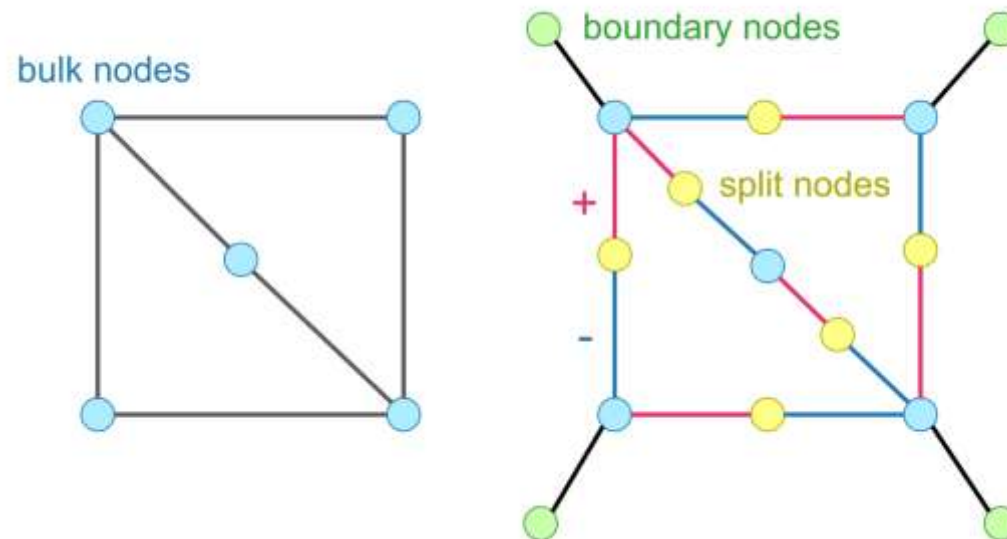


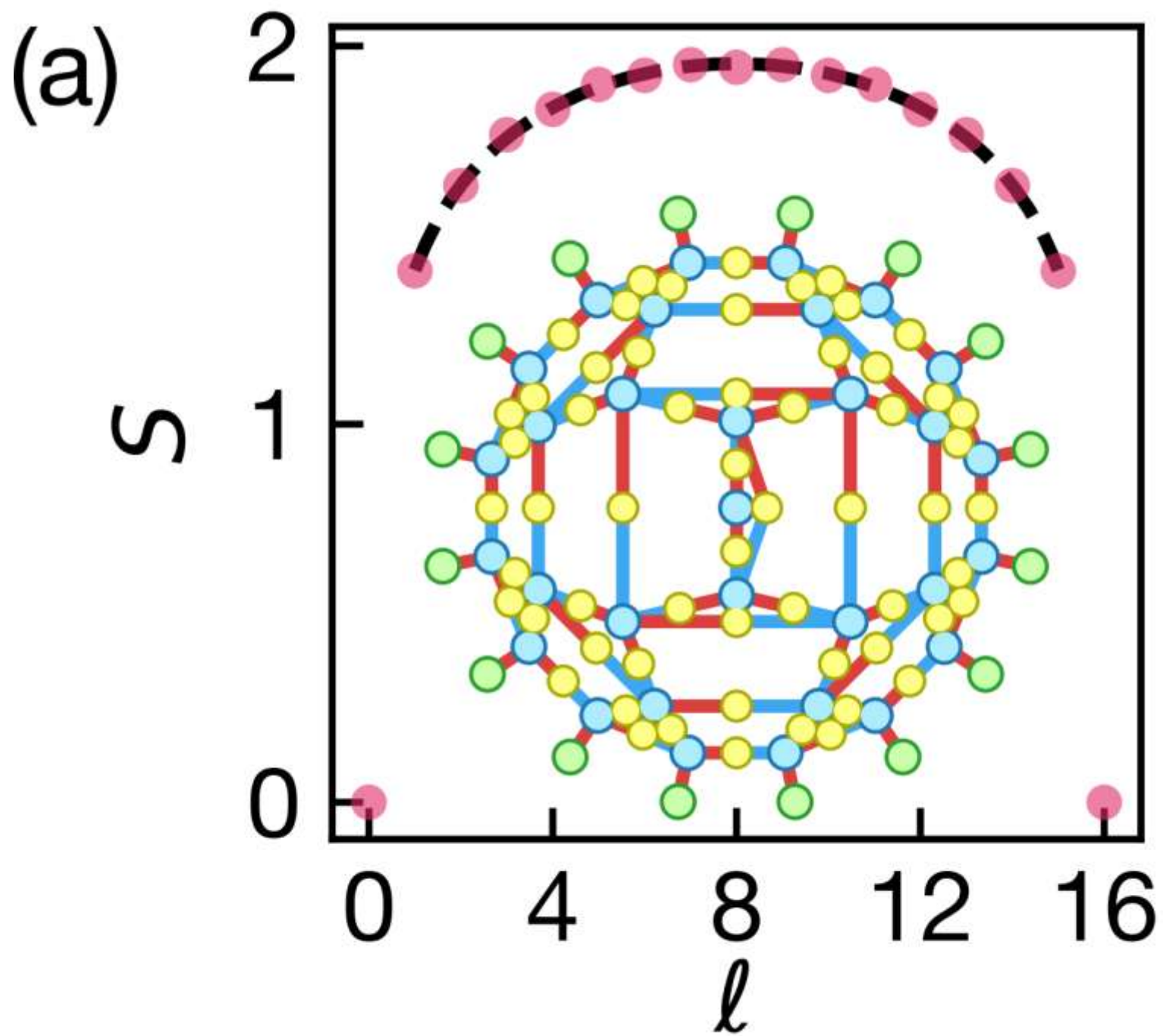
(c)

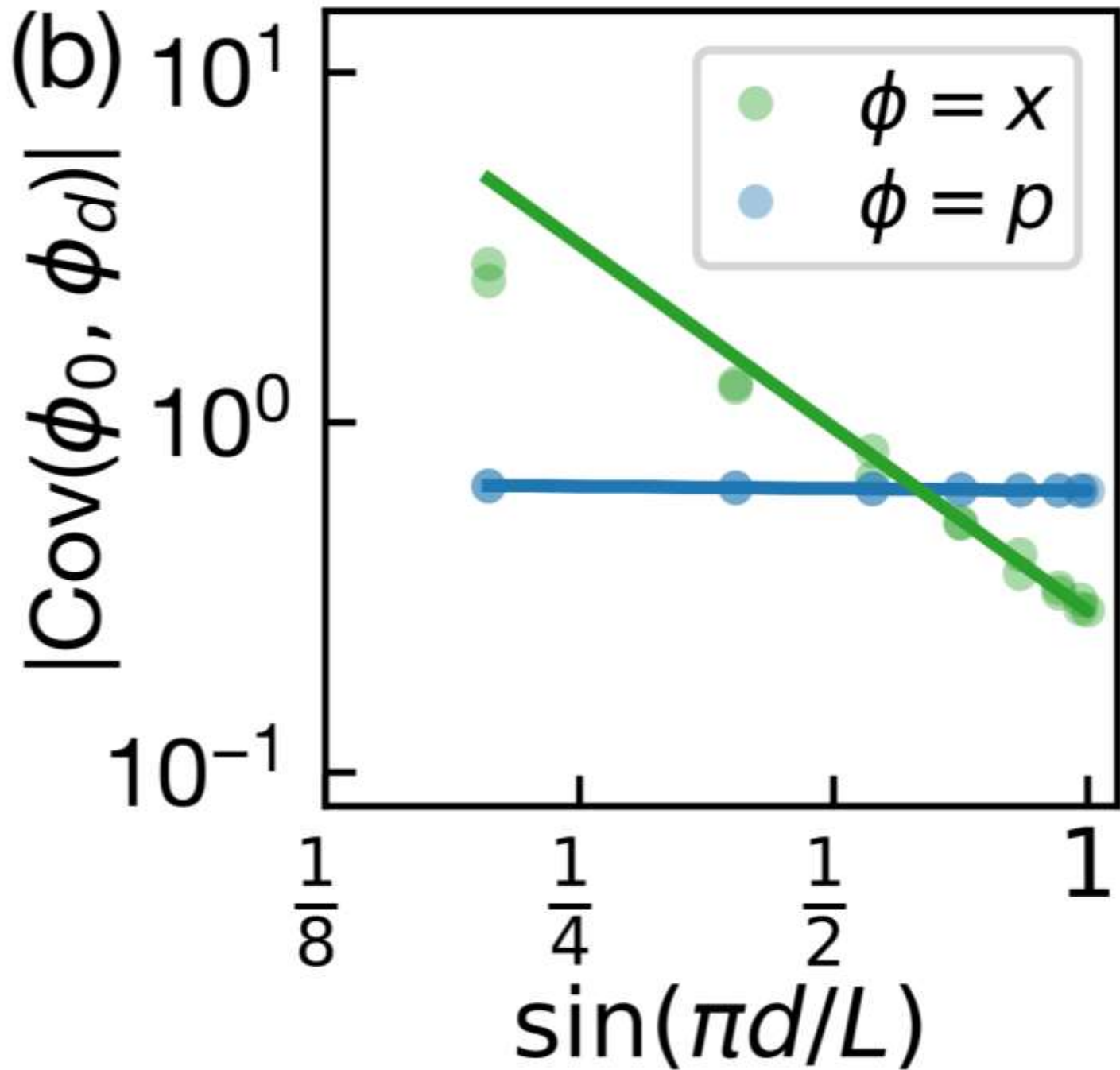


Example 3: decorated hyperbolic disk

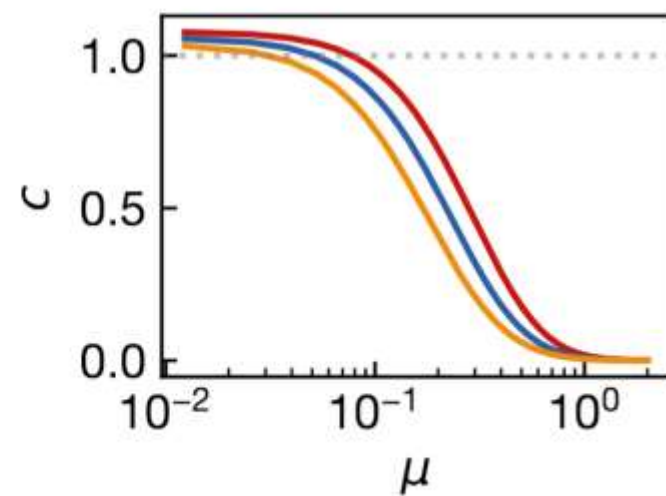
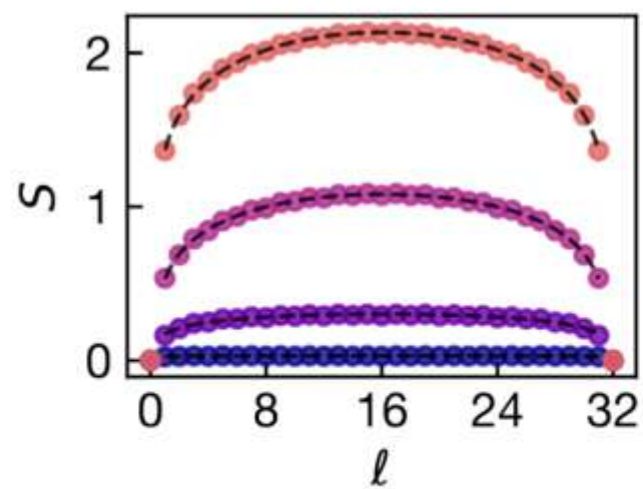
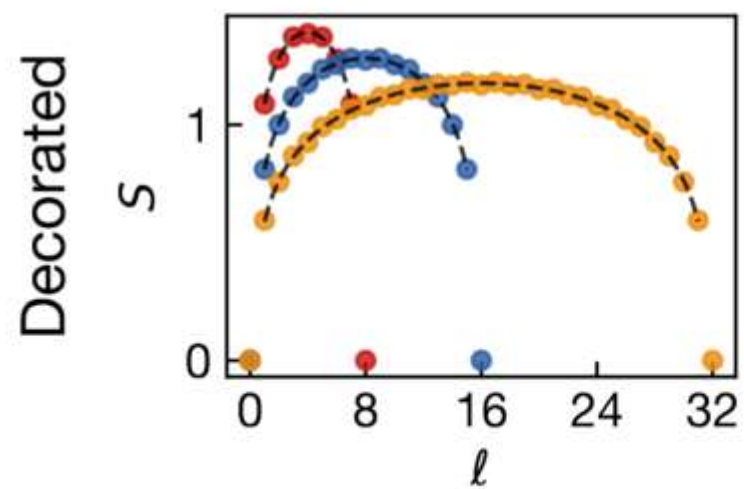
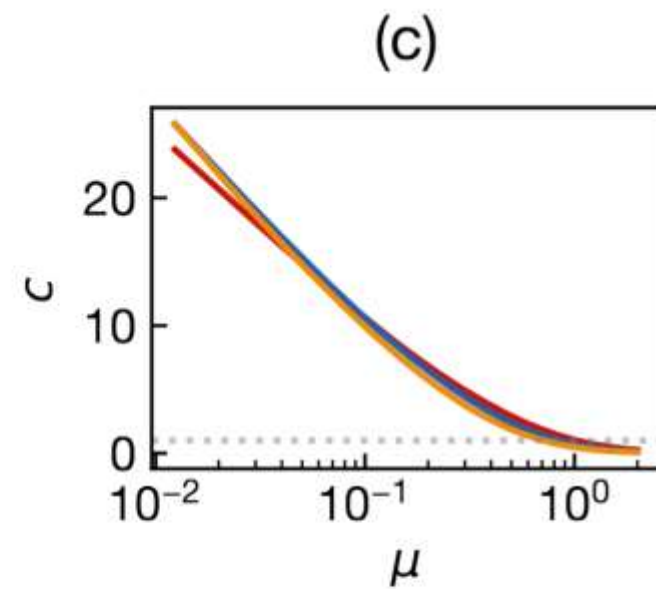
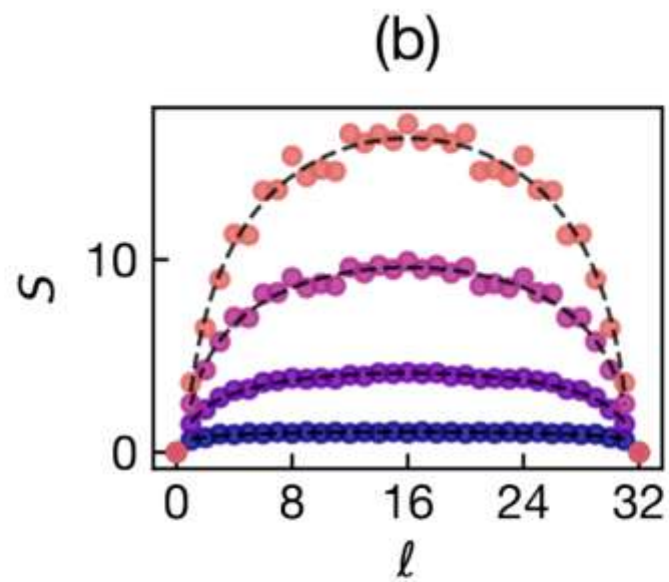
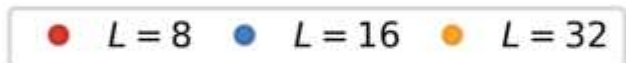
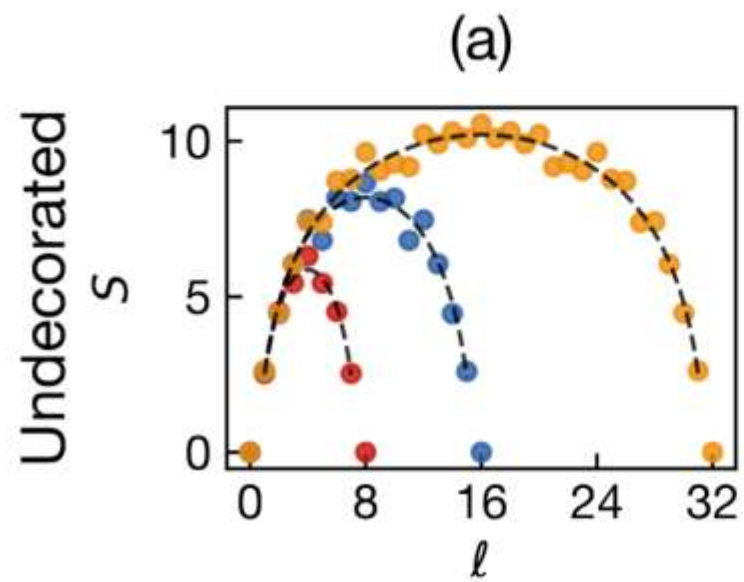
- We've achieved RT-like entanglement entropy and matched CFT Renyi entropy, at least for single intervals in the ground state
- We also wanted to obtain power-law correlations, which turns out to be possible using a decoration procedure







Scaling dimensions ($\approx 0,1$), central charge (≈ 1), and Wick's theorem are close to our expectations for a free boson CFT ground state. [Did we build this CFT state at long wavelengths?](#)

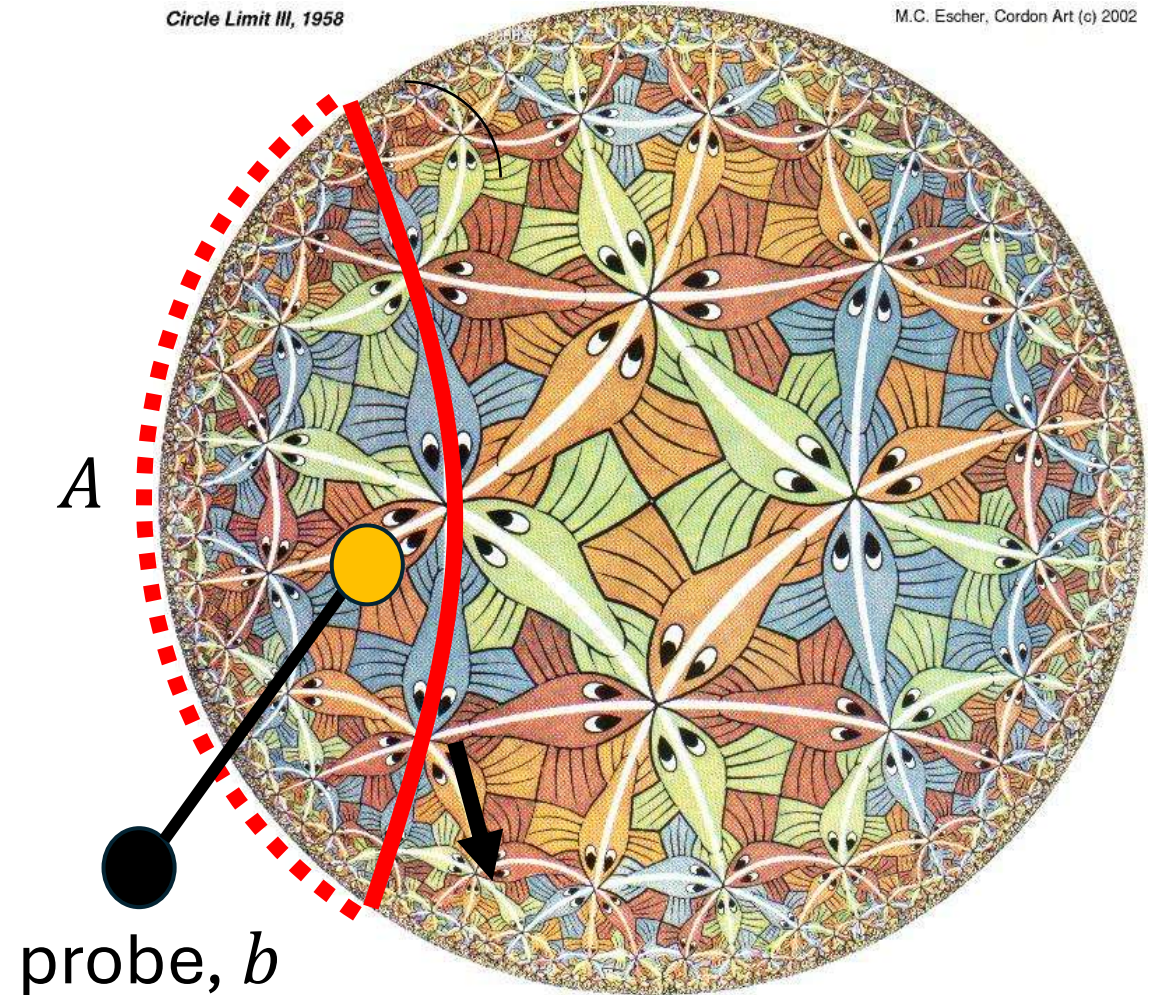


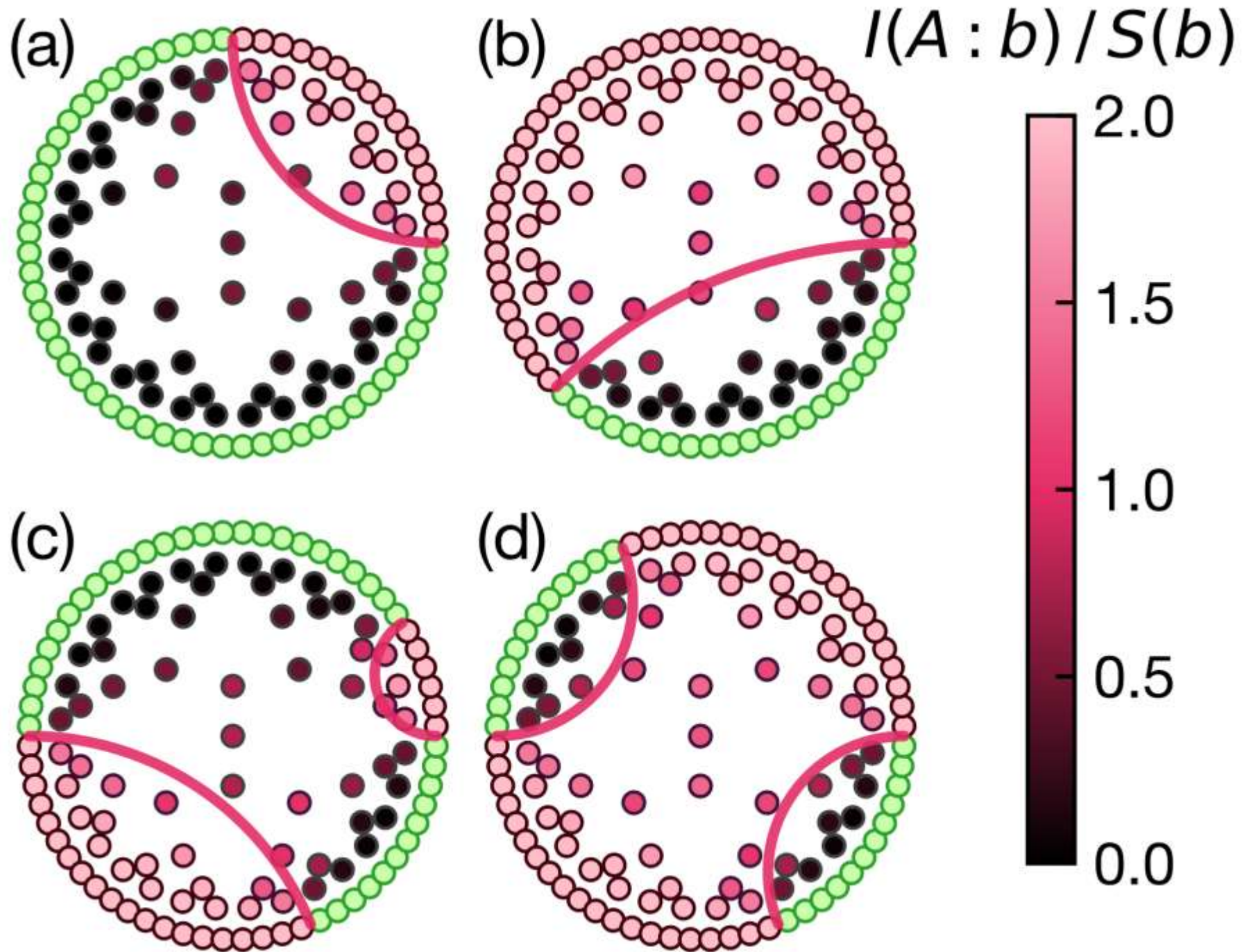
Bulk reconstruction

- We'd like to directly detect the RT surface from data
- Add one extra site, the probe, which is unmeasured and directly connects to a chosen bulk site ●
- AdS/CFT: $I(A: b)/S(b) = 2$ if probe connects inside the RT surface of A and $= 0$ if it connects outside

$$I(A: b) = S(A) + S(b) - S(Ab)$$

mutual information





Outlook

There are a lot of potential directions, including:

- We want to think about adding non-Gaussianity and dealing with post-selection [e.g. backcreation [Cao et al. 2403.07056?](#)]
- Experimental implementation is also crucial, and I think the quantum extremal surface formalism will help us model the effects of imperfections, e.g. incomplete bulk measurements
- The wormholes can be viewed as TFD states of some quadratic Hamiltonian, so we can study teleportation dynamics ([WIP Allison Powell, Brian Khor](#))

Perturbing the hyperbolic disk $g_{ij}^{(0)} \mapsto g_{ij} = g_{ij}^{(0)} + h_{ij}$

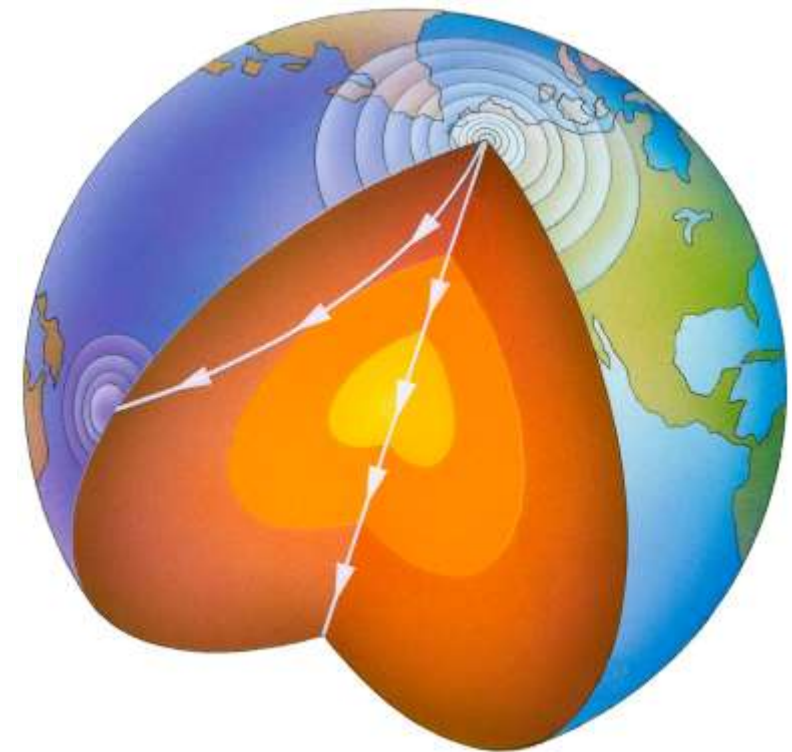
- Tensor Radon transform: $R_2[h_{ij}](\gamma_A) \equiv \int_{\gamma_A} h_{ij} \dot{\gamma}_A^i(s) \dot{\gamma}_A^j(s) ds$

- Idea: invert transform to determine metric perturbation from entanglement perturbation
- Must be careful about gauge-invariance

$$R_2[\nabla_i \xi_j + \nabla_j \xi_i] = 0$$

- Related problems appear in geophysics, where one wants to learn about the properties of the Earth's interior from seismic wave timing data

[Tang-Cao-Qi-S 2007.00004]



[from DK Find Out website]

Algorithm

- Discretize the bulk geometry into tiles with piecewise constant metric
- Radon transform takes the form of a linear map from space of metric perturbations to space of entanglement perturbations

$$2\Delta L(\gamma_A) = R_2[h_{ij}] \approx \sum_{\mathcal{T}} W^{ij}(\mathcal{T}, \gamma_A) h_{ij}(\mathcal{T})$$

- Invert this map within a gauge fixed sector of metric perturbations

$$\min_{\mathbf{h}} \|\mathbf{W}\mathbf{h} - \mathbf{b}\|, \quad \mathbf{h}_* \text{ minimizer}$$

subject to $\mathbf{C}\mathbf{h} = \mathbf{0}$. gauge fixing constraints

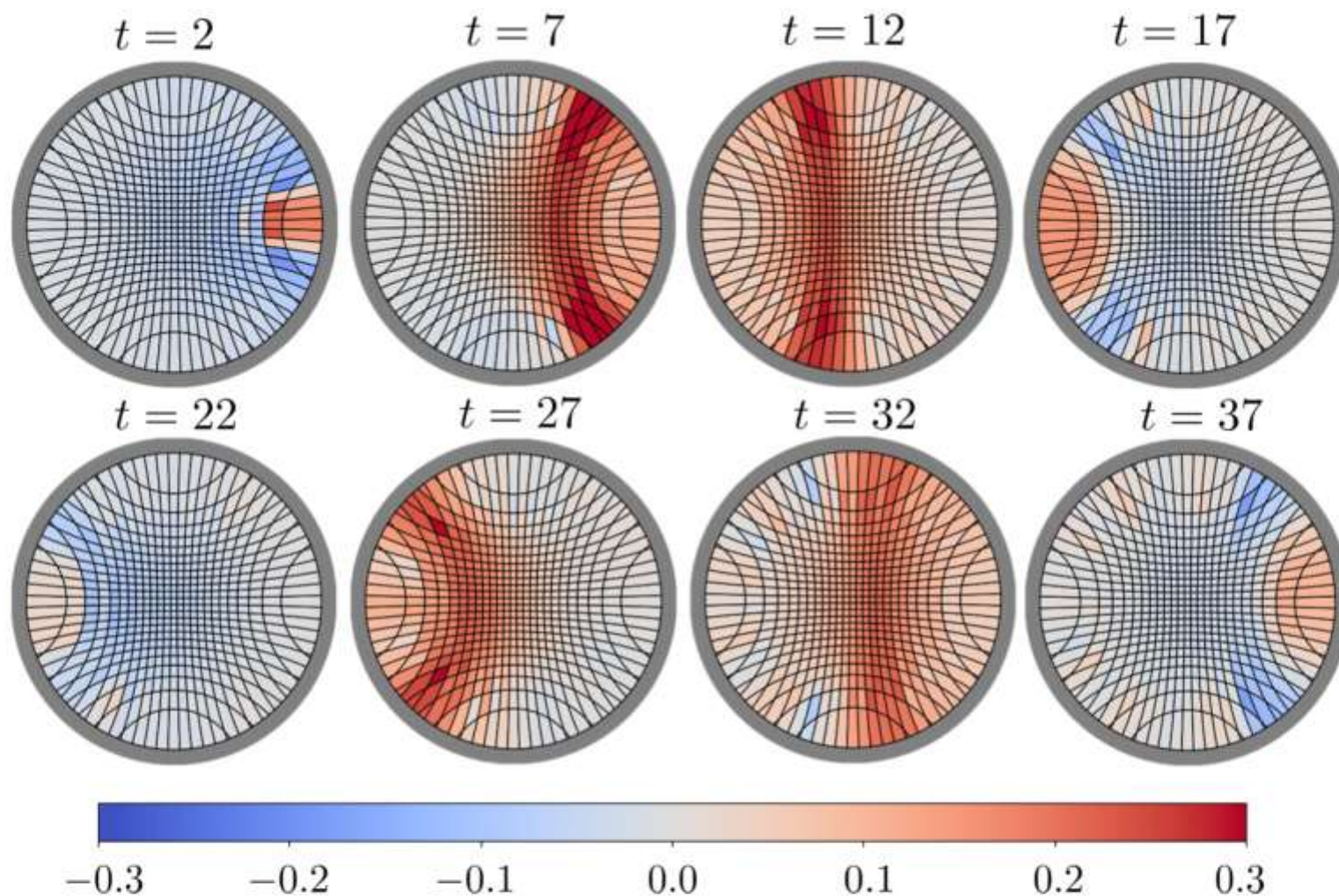


Figure 6: Quenched time dynamics of the reconstructed geometry corresponding to a single mass deformation along the x -axis. The boundary data is pre-processed by smoothing to remove small-scale details (see Figure 5 for the corresponding unsmoothed reconstruction). Only the dominant component h_{11} is shown here. The relative error is $0.03 \lesssim \mathcal{E}_{\text{bdy}} \lesssim 0.1$.

Gravitational dynamics?

- In 2014, Van Raamsdonk and I showed that linearized Einstein's equations and the equivalence principle emerged from:
 - (i) the RT formula + quantum corrections
 - (ii) CFT symmetries
- The setting was the same as the just discussed problem of metric reconstruction, namely perturbations around AdS
- We've also seen RT and CFT-like behaviors earlier in the talk
- Our protocol for building holographic entanglement by measurement seems like an excellent starting point for exploring these issues in the lab! How much non-Gaussianity do we need?

